PROJECT REPORT & SYNTHESIS

Submitted by- SARIF AHMAD

a) Design a Synchronous FIFO by using the Verilog HDL.

b) simulate it using VCS and Verdi.

c) Verify it through the testbench.

d) Synthesis done by Synopsys Design compiler tool(32nm) and created the Mapping netlist file by SDC tool.

FIFO (FIRST IN FIRST OUT): -

A **FIFO** circuit is a special type of data buffer (**storage element**) used in digital systems designed to manage data flow. The name FIFO stands for **first-in-first-out** and means that the data written (entered) into the buffer first comes out of it first similar to a queue provides sequential data flow between two systems.

Data is written starting at location 0 and then stored sequentially in 1,2, 3……. upto FIFO_DEPTH-1, as data arrives. When performing a read operation from fifo, the data is read from location 0 followed sequentially by location 1,2, …. upto FIFO_DEPTH-1. So, we can observe that the data which entered first is read out first.

Application of FIFOs-

- Data streaming applications, where data needs to be processed in the same order as it arrives. FIFO ensure that the data flows smoothly from entry to exit without changing the sequence which is crucial for streaming (**data transfer**).
- Data **Buffering** (a **temporary storage** element) is needed when there is mismatch in processing speeds of the module1 and module2 devices. Module1 is the device generating or writing the data and Module2 is the device reading or processing the data. When Module1 generates the data faster then the Module2 can process it, a FIFO buffers helps temporarily store the data to ensure efficient transfer between the two devices. This way we can avoid any risk of data loss or overflow.
- **Data synchronization** between systems operating at different clock domains. In real world applications, especially on SOC designs, multiple blocks on the chip often operate with separate clocks. FIFO circuits help ensure smooth and accurate data transfers between these blocks. By carefully managing the timing differences, FIFO circuits prevent data loss and corruption during transfer.

Types of FIFOs-

- **Synchronous FIFO** is a data buffer circuit, in which there is a **single clock** signal for both data write and data read. The synchronization between the device is simple, since they shared same timing.
- **Asynchronous FIFO** is a FIFO circuit where the data values are written and read back to the FIFO at **different frequency** rates (different clocks). This design allows reliable data transfer even when the clock between the devices are mismatched.

Synchronous FIFO:

A **Synchronous FIFO** (First-In First-Out) is a type of data buffer used in digital systems that read and write operations occur under the same clock domains. It requires control logic for full, empty and almost full/empty flag, overflow and underflow status signals. The read and write pointers are updated synchronously with the clock.

FIFO_DEPTH-

The depth of a FIFO refers to the **number of the storage locations available** for holding data before the buffer is full. It is crucial for ensuring proper data flow without overflow or underflow conditions. In other terms, we can say that the number of slots or rows in FIFO memory is called the depth of the FIFO.

DATA_WIDTH OF FIFO-

The number of bits that can be stored in each slot or row of FIFO is called the width of the FIFO (also called word length of FIFO).

WORKING PRINCIPLE: -

- Data is written into the FIFO on a **same clock edge till FIFO full**.
- Data is read from the FIFO on a **same clock edge till FIFO empty.**

Functionality and Parameters of synchronous FIFO: -

Here, **16x8 Synchronous fifo means** FIFO_DEPTH is 16 storage locations (capacity of FIFO) and DATA_WIDTH is **8bits** means each data is of 8-bits**.**

Memory (memory 16x8): is used to temporarily store the data within fifo.

FIFO full **Condition:** Flag to indicate if the buffer is full (full=1).

FIFO empty **Condition:** Flag to indicate if the buffer is empty (empty=1).

Write Pointer (wr_ptr**):** Points to the memory location in FIFO from where data will be written upto the reaches the maximum capacity, here its value is 15.

Read Pointer (rd_ptr)**:** Points to the memory location in FIFO from where data is to be read.

Overflow Condition (wr_en=1, full=1)**:**

When fifo memory is full but we want to write data again and again in a full FIFO. So, overflow signal will show value 1 means we are overriding the data in fifo. These flags are used to prevent the overflow condition.

Underflow Condition (rd_en=1, empty=1)**:**

When fifo memory is empty but we want to read data again and again in an empty FIFO. So, underflow signal will show value 1. These flags are used to prevent the underflow condition in fifo.

Write Operation: Data is written at the location in memory array specified by the write pointer, and the pointer increments after each write (wr_en=1, rd_en=0).

Read Operation: Data is read from the location in memory array specified by the read pointer, and the read pointer increments after each read the data (rd_en=1, wr_en=0).

Write/Read Operation: When wr_en=1, rd_en=1, simultaneously write and read happen at the same clock cycle. The data count remains unchanged.

Counter: count varies from 0 to 15 and counts the number of elements in FIFO.

clk: Positive edge triggering for synchronizing the operations.

rst_n: Active low reset signal to initialize the FIFO (empty=1).

data_in: input data to be written in FIFO.

wr_en: write enable signal. (wr_en=1) means data is written.

rd_en: read enable signal. (rd_en=1) means data is read.

data_out: output data read from the fifo.

Design

Here, we are designing a **16x8 Synchronous fifo** using macros and `define compiler directive. The design includes **input pins** like clk, rst_n, wr_en, rd_en, and an **8-bit data_in,** along with **output pins** like the **8-bit data_out**, and status signals: underflow, overflow, full, empty.

The FIFO functionality is implemented using four procedural blocks: One procedural block for reading and writing operations of FIFO, the second procedural block for read pointers

and write pointers movement, the third procedural block for FIFO status like full and empty, the fourth procedural block for generating flags based on FIFO status. Finally, the design will be will **verified through testbenches** that cover at all possible scenarios.

Design code

```
//Design the 16x8 synchronous FIFO

`define DATA_WIDTH 8

`define FIFO_DEPTH 16

`define ADDR_WIDTH 4


module Sync_fifo_16x8 (data_out, underflow, overflow, full, empty, data_in, clk, rst_n, rd_en, wr_en);


input logic clk, rst_n, rd_en, wr_en;

input logic [`DATA_WIDTH -1:0] data_in;

output logic [`DATA_WIDTH -1:0] data_out;

output logic underflow, overflow, full, empty;


//FIFO memory and pointers

logic [`DATA_WIDTH-1:0] memory [`FIFO_DEPTH-1:0];

logic [`ADDR_WIDTH-1:0] wr_ptr_reg, rd_ptr_reg;

logic [`ADDR_WIDTH:0] count_reg;


logic [`DATA_WIDTH -1:0] data_in_reg, data_out_reg;

logic wr_en_reg, rd_en_reg;


always_ff @ (posedge clk or negedge rst_n) begin

        if (! rst_n) begin
```

```
                data_in_reg <=0;
                wr_en_reg  <=0;
                rd_en_reg  <=0;
        end
        else begin
                data_in_reg <= data_in;
                wr_en_reg  <=wr_en;
                rd_en_reg  <=rd_en;
        end
end
always_ff @ (posedge clk or negedge rst_n) begin
        if (! rst_n) begin
                wr_ptr_reg <=0;
                rd_ptr_reg <=0;
                count_reg  <=0;
               data_out_reg <=0;
        end
        else begin
                if ((wr_en_reg==1'b1) && (full==1'b0)) begin
                   memory[wr_ptr_reg] <= data_in_reg;
                   wr_ptr_reg <= (wr_ptr_reg==`FIFO_DEPTH-1) ? 0 : wr_ptr_reg +1;
                end
                if ((rd_en_reg==1'b1) && (empty==1'b0)) begin
                   data_out_reg <= memory[rd_ptr_reg];
                   rd_ptr_reg <= (rd_ptr_reg==`FIFO_DEPTH-1) ? 0 : rd_ptr_reg +1;
                end
                case ({(wr_en_reg==1'b1) && (full==1'b0), (rd_en_reg==1'b1) &&
(empty==1'b0)})
```

```
                2'b10: count_reg <= count_reg +1;
                2'b01: count_reg <= count_reg -1;
                2'b11: count_reg <= count_reg;
                default: ;
            endcase
        end
end

always_comb begin
        full = (count_reg == `FIFO_DEPTH);
        empty = (count_reg == 0);
end

//overflow and underflow flag status
always_ff @ (posedge clk or negedge rst_n) begin
        if (! rst_n) begin
                overflow <= 0;
                underflow <= 0;
        end
        else begin
                overflow <= ((wr_en_reg==1'b1) && (full==1'b1));
                underflow <= ((rd_en_reg==1'b1) && (empty==1'b1));
        end
end
always_ff @ (posedge clk or negedge rst_n) begin
        if (! rst_n)
                data_out <= 0;
```

```
        else
                data_out <= data_out_reg;
end
endmodule
```

Testbench

```
// Testbench of 16x8 synchronous FIFO
`define DATA_WIDTH 8
`define FIFO_DEPTH 16
`define ADDR_WIDTH 4

module Sync_fifo_tb ();
reg clk, rst_n, rd_en, wr_en;
reg [`DATA_WIDTH -1:0] data_in;
wire [`DATA_WIDTH -1:0] data_out;
wire underflow, overflow, full, empty;

Sync_fifo_16x8 dut (.clk (clk), .rst_n(rst_n), .rd_en(rd_en), .wr_en(wr_en), .data_in(data_in), .data_out(data_out), .underflow(underflow), .overflow(overflow), .full(full),.empty(empty));

initial clk=0;
always #5 clk=~clk; //clock generation
integer i;

initial begin
        $monitor ("Time=%0t for rst_n=%b wr_en=%b rd_en=%b data_in=%d we get data_out=%d underflow=%b overflow=%b full=%b empty=%b", $time, rst_n, wr_en, rd_en, data_in, data_out, underflow, overflow, full, empty);
```

```
        rst_n=0; wr_en=0; rd_en=0; data_in=0;
#10 rst_n=1;

//write in FIFO until full completely
wr_en=1;
for (i=0; I < `FIFO_DEPTH; i=i+1) begin
        data_in = i;
        #10;
end
//Try to overflow the data in fifo
for (i=0; i< 5; i=i+1) begin
      data_in = 50+i;
      #10;
end

wr_en=0;
//Read all data until empty
rd_en=1;
for (i=0; i< `FIFO_DEPTH; i=i+1) begin
      #10;
end
//try to read more data in underflow condition
for (i=0; i< 5; i=i+1) begin
      #10;
end
rd_en=0;
```

```
//simultaneously read/write operation
wr_en=1; rd_en=1;
for (i=0; I < `FIFO_DEPTH; i=i+1) begin
    data_in = 200+i;
    #10;
end
#10 $display ("Time=%0t for rst_n=%b wr_en=%b rd_en=%b data_in=%d we get data_out=%d underflow=%b overflow=%b full=%b empty=%b", $time, rst_n, wr_en, rd_en, data_in, data_out, underflow, overflow, full, empty);
wr_en=0; rd_en=0;

$display ("All tests have been passed");
#10 $finish ();
end
initial begin
    $fsdbDumpfile("dump.fsdb");
    $fsdbDumpvars(0,Sync_fifo_tb);
end
endmodule
```

Run command

```
vcs -sverilog sync_fifo.sv sync_fifo_tb.sv -kdb -lca -debug_access+all -full64
```

Simulation Result

Output

```
Time=0 for rst_n=0 wr_en=0 rd_en=0 data_in=  0 we get data_out=  0 underflow=0 overflow=0 full=0 empty=1

Time=10 for rst_n=1 wr_en=1 rd_en=0 data_in=  0 we get data_out=  0 underflow=0 overflow=0 full=0 empty=1
```

Time=20 for rst_n=1 wr_en=1 rd_en=0 data_in= 1 we get data_out= 0 underflow=0 overflow=0 full=0 empty=1

Time=25 for rst_n=1 wr_en=1 rd_en=0 data_in= 1 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=30 for rst_n=1 wr_en=1 rd_en=0 data_in= 2 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=40 for rst_n=1 wr_en=1 rd_en=0 data_in= 3 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=50 for rst_n=1 wr_en=1 rd_en=0 data_in= 4 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=60 for rst_n=1 wr_en=1 rd_en=0 data_in= 5 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=70 for rst_n=1 wr_en=1 rd_en=0 data_in= 6 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=80 for rst_n=1 wr_en=1 rd_en=0 data_in= 7 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=90 for rst_n=1 wr_en=1 rd_en=0 data_in= 8 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=100 for rst_n=1 wr_en=1 rd_en=0 data_in= 9 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=110 for rst_n=1 wr_en=1 rd_en=0 data_in= 10 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=120 for rst_n=1 wr_en=1 rd_en=0 data_in= 11 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=130 for rst_n=1 wr_en=1 rd_en=0 data_in= 12 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=140 for rst_n=1 wr_en=1 rd_en=0 data_in= 13 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=150 for rst_n=1 wr_en=1 rd_en=0 data_in= 14 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=160 for rst_n=1 wr_en=1 rd_en=0 data_in= 15 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=170 for rst_n=1 wr_en=1 rd_en=0 data_in= 50 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=175 for rst_n=1 wr_en=1 rd_en=0 data_in= 50 we get data_out= 0 underflow=0 overflow=0 full=1 empty=0

Time=180 for rst_n=1 wr_en=1 rd_en=0 data_in= 51 we get data_out= 0 underflow=0 overflow=0 full=1 empty=0

Time=185 for rst_n=1 wr_en=1 rd_en=0 data_in= 51 we get data_out= 0 underflow=0 overflow=1 full=1 empty=0

Time=190 for rst_n=1 wr_en=1 rd_en=0 data_in= 52 we get data_out= 0 underflow=0 overflow=1 full=1 empty=0

Time=200 for rst_n=1 wr_en=1 rd_en=0 data_in= 53 we get data_out= 0 underflow=0 overflow=1 full=1 empty=0

Time=210 for rst_n=1 wr_en=1 rd_en=0 data_in= 54 we get data_out= 0 underflow=0 overflow=1 full=1 empty=0

Time=220 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 0 underflow=0 overflow=1 full=1 empty=0

Time=235 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 0 underflow=0 overflow=0 full=0 empty=0

Time=255 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 1 underflow=0 overflow=0 full=0 empty=0

Time=265 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 2 underflow=0 overflow=0 full=0 empty=0

Time=275 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 3 underflow=0 overflow=0 full=0 empty=0

Time=285 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 4 underflow=0 overflow=0 full=0 empty=0

Time=295 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 5 underflow=0 overflow=0 full=0 empty=0

Time=305 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 6 underflow=0 overflow=0 full=0 empty=0

Time=315 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 7 underflow=0 overflow=0 full=0 empty=0

Time=325 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 8 underflow=0 overflow=0 full=0 empty=0

Time=335 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 9 underflow=0 overflow=0 full=0 empty=0

Time=345 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 10 underflow=0 overflow=0 full=0 empty=0

Time=355 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 11 underflow=0 overflow=0 full=0 empty=0

Time=365 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 12 underflow=0 overflow=0 full=0 empty=0

Time=375 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 13 underflow=0 overflow=0 full=0 empty=0

Time=385 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 14 underflow=0 overflow=0 full=0 empty=1

Time=395 for rst_n=1 wr_en=0 rd_en=1 data_in= 54 we get data_out= 15 underflow=1 overflow=0 full=0 empty=1

Time=430 for rst_n=1 wr_en=1 rd_en=1 data_in=200 we get data_out= 15 underflow=1 overflow=0 full=0 empty=1

Time=440 for rst_n=1 wr_en=1 rd_en=1 data_in=201 we get data_out= 15 underflow=1 overflow=0 full=0 empty=1

Time=445 for rst_n=1 wr_en=1 rd_en=1 data_in=201 we get data_out= 15 underflow=1 overflow=0 full=0 empty=0

Time=450 for rst_n=1 wr_en=1 rd_en=1 data_in=202 we get data_out= 15 underflow=1 overflow=0 full=0 empty=0

Time=455 for rst_n=1 wr_en=1 rd_en=1 data_in=202 we get data_out= 15 underflow=0 overflow=0 full=0 empty=0

Time=460 for rst_n=1 wr_en=1 rd_en=1 data_in=203 we get data_out= 15 underflow=0 overflow=0 full=0 empty=0

Time=465 for rst_n=1 wr_en=1 rd_en=1 data_in=203 we get data_out=200 underflow=0 overflow=0 full=0 empty=0

Time=470 for rst_n=1 wr_en=1 rd_en=1 data_in=204 we get data_out=200 underflow=0 overflow=0 full=0 empty=0

Time=475 for rst_n=1 wr_en=1 rd_en=1 data_in=204 we get data_out=201 underflow=0 overflow=0 full=0 empty=0

Time=480 for rst_n=1 wr_en=1 rd_en=1 data_in=205 we get data_out=201 underflow=0 overflow=0 full=0 empty=0

Time=485 for rst_n=1 wr_en=1 rd_en=1 data_in=205 we get data_out=202 underflow=0 overflow=0 full=0 empty=0

Time=490 for rst_n=1 wr_en=1 rd_en=1 data_in=206 we get data_out=202 underflow=0 overflow=0 full=0 empty=0

Time=495 for rst_n=1 wr_en=1 rd_en=1 data_in=206 we get data_out=203 underflow=0 overflow=0 full=0 empty=0

Time=500 for rst_n=1 wr_en=1 rd_en=1 data_in=207 we get data_out=203 underflow=0 overflow=0 full=0 empty=0

Time=505 for rst_n=1 wr_en=1 rd_en=1 data_in=207 we get data_out=204 underflow=0 overflow=0 full=0 empty=0

Time=510 for rst_n=1 wr_en=1 rd_en=1 data_in=208 we get data_out=204 underflow=0 overflow=0 full=0 empty=0

Time=515 for rst_n=1 wr_en=1 rd_en=1 data_in=208 we get data_out=205 underflow=0 overflow=0 full=0 empty=0

Time=520 for rst_n=1 wr_en=1 rd_en=1 data_in=209 we get data_out=205 underflow=0 overflow=0 full=0 empty=0

Time=525 for rst_n=1 wr_en=1 rd_en=1 data_in=209 we get data_out=206 underflow=0 overflow=0 full=0 empty=0

Time=530 for rst_n=1 wr_en=1 rd_en=1 data_in=210 we get data_out=206 underflow=0 overflow=0 full=0 empty=0

Time=535 for rst_n=1 wr_en=1 rd_en=1 data_in=210 we get data_out=207 underflow=0 overflow=0 full=0 empty=0

Time=540 for rst_n=1 wr_en=1 rd_en=1 data_in=211 we get data_out=207 underflow=0 overflow=0 full=0 empty=0

Time=545 for rst_n=1 wr_en=1 rd_en=1 data_in=211 we get data_out=208 underflow=0 overflow=0 full=0 empty=0

Time=550 for rst_n=1 wr_en=1 rd_en=1 data_in=212 we get data_out=208 underflow=0 overflow=0 full=0 empty=0

Time=555 for rst_n=1 wr_en=1 rd_en=1 data_in=212 we get data_out=209 underflow=0 overflow=0 full=0 empty=0

Time=560 for rst_n=1 wr_en=1 rd_en=1 data_in=213 we get data_out=209 underflow=0 overflow=0 full=0 empty=0

Time=565 for rst_n=1 wr_en=1 rd_en=1 data_in=213 we get data_out=210 underflow=0 overflow=0 full=0 empty=0

Time=570 for rst_n=1 wr_en=1 rd_en=1 data_in=214 we get data_out=210 underflow=0 overflow=0 full=0 empty=0

Time=575 for rst_n=1 wr_en=1 rd_en=1 data_in=214 we get data_out=211 underflow=0 overflow=0 full=0 empty=0

Time=580 for rst_n=1 wr_en=1 rd_en=1 data_in=215 we get data_out=211 underflow=0 overflow=0 full=0 empty=0

Time=585 for rst_n=1 wr_en=1 rd_en=1 data_in=215 we get data_out=212 underflow=0 overflow=0 full=0 empty=0

Time=595 for rst_n=1 wr_en=1 rd_en=1 data_in=215 we get data_out=213 underflow=0 overflow=0 full=0 empty=0

Time=600 for rst_n=1 wr_en=1 rd_en=1 data_in=215 we get data_out=213 underflow=0 overflow=0 full=0 empty=0

All tests have been passed

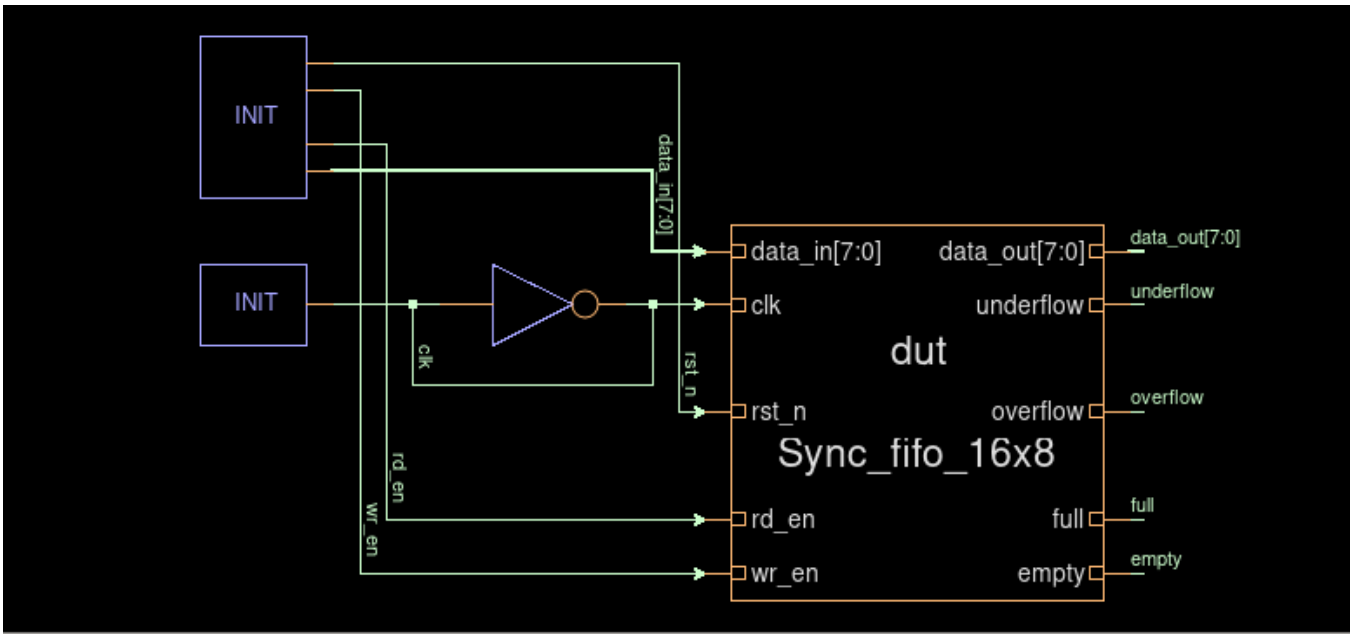
Time=600 for rst_n=1 wr_en=0 rd_en=0 data_in=215 we get data_out=213 underflow=0 overflow=0 full=0 empty=0

Time=605 for rst_n=1 wr_en=0 rd_en=0 data_in=215 we get data_out=214 underflow=0 overflow=0 full=0 empty=0

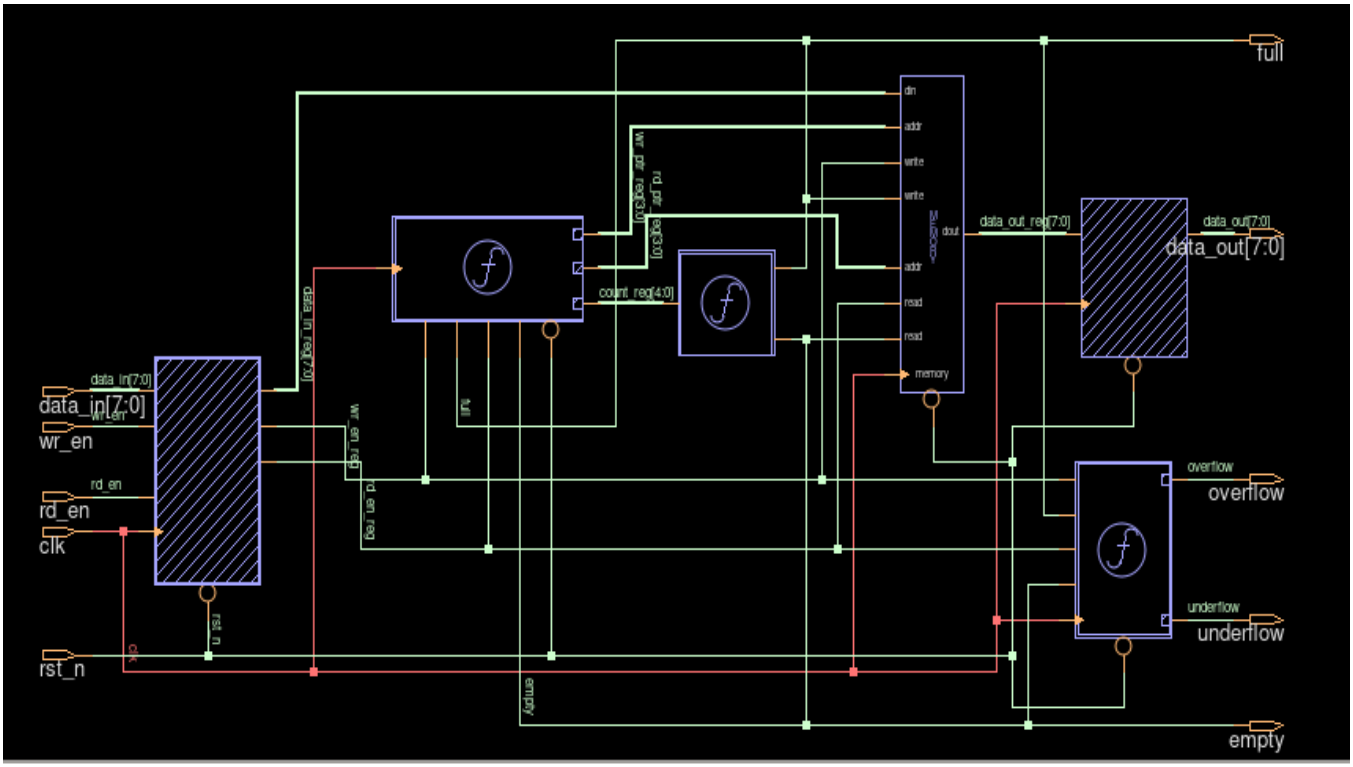
$finish called from file "sync_fifo_tb.sv", line 63.

$finish at simulation time 610

Schematic Design

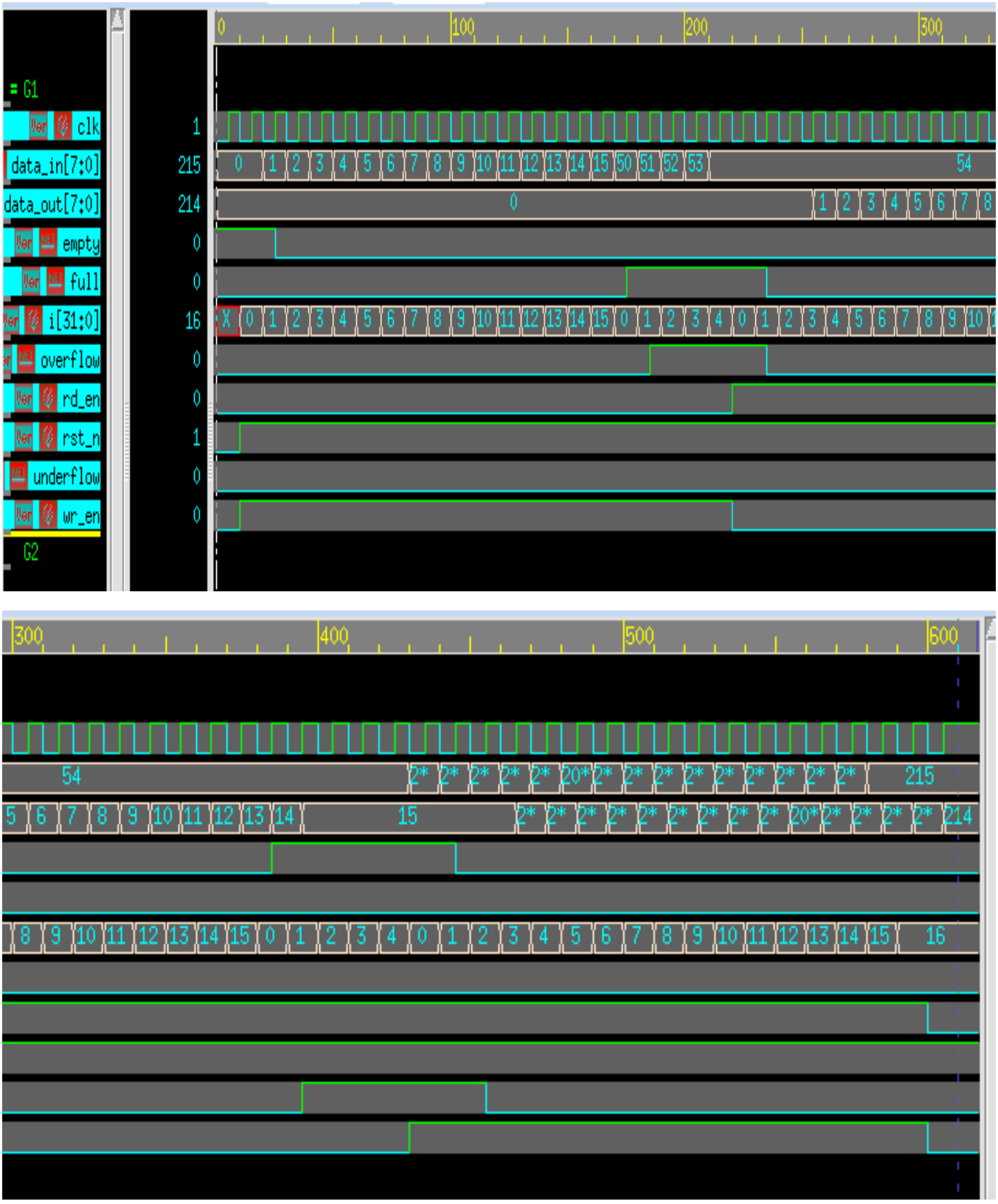


(Schematic View of 16x8 Synchronous fifo - by VCS and Verdi tool)



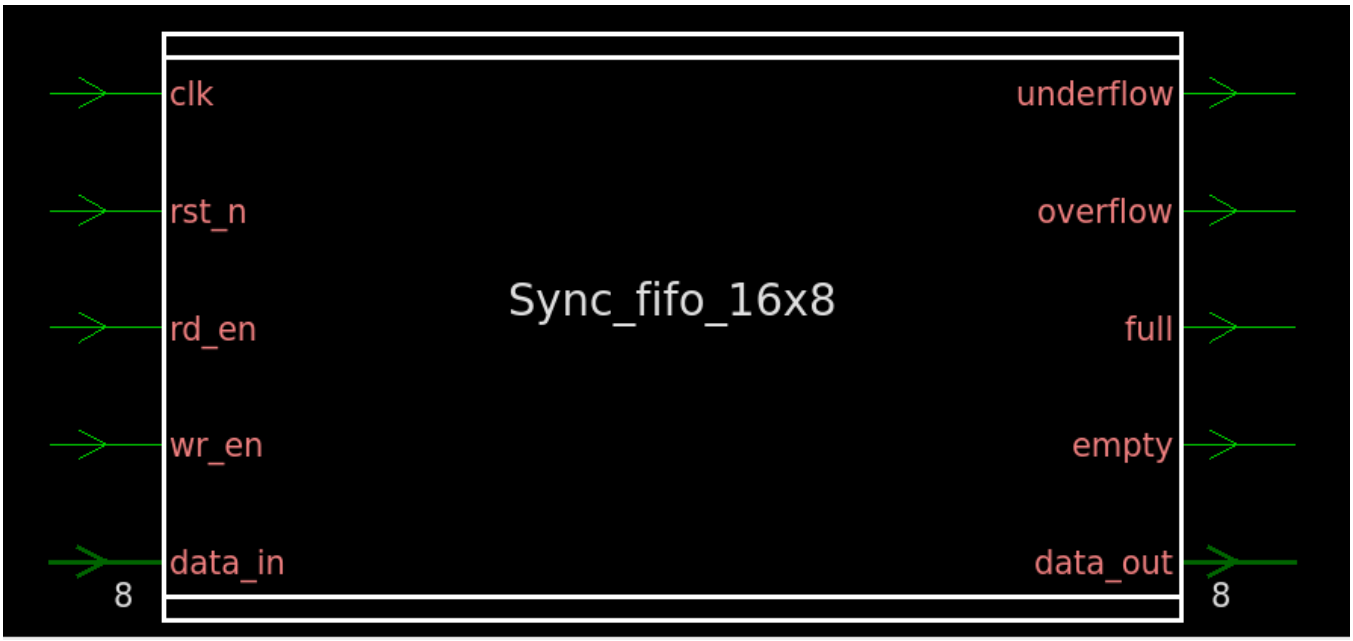
(This is the internal schematic view of synthesizable circuits of 16x8 synchronous FIFO)

Waveform-

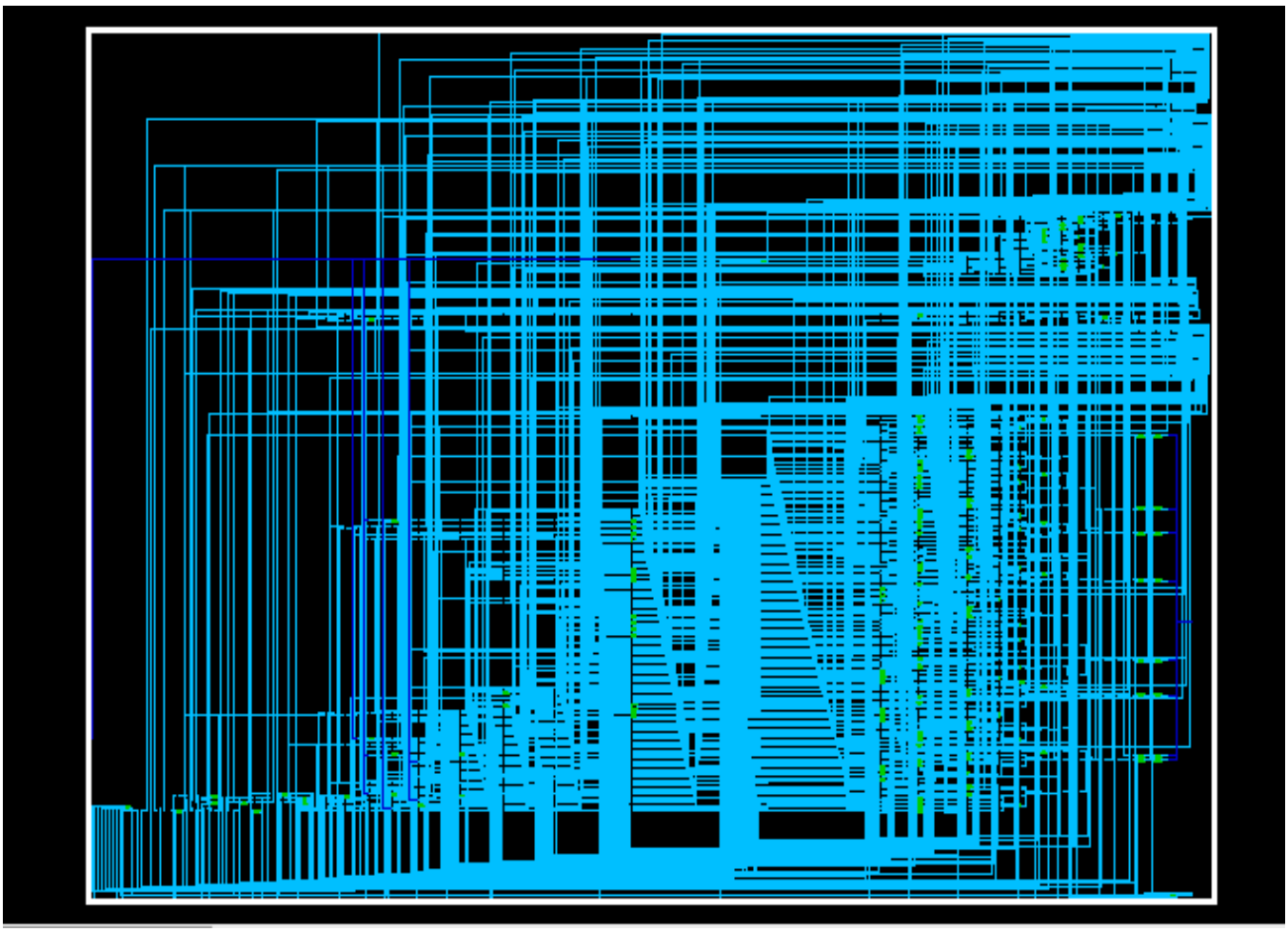


(Output waveform by using the Vcs and Verdi toll)

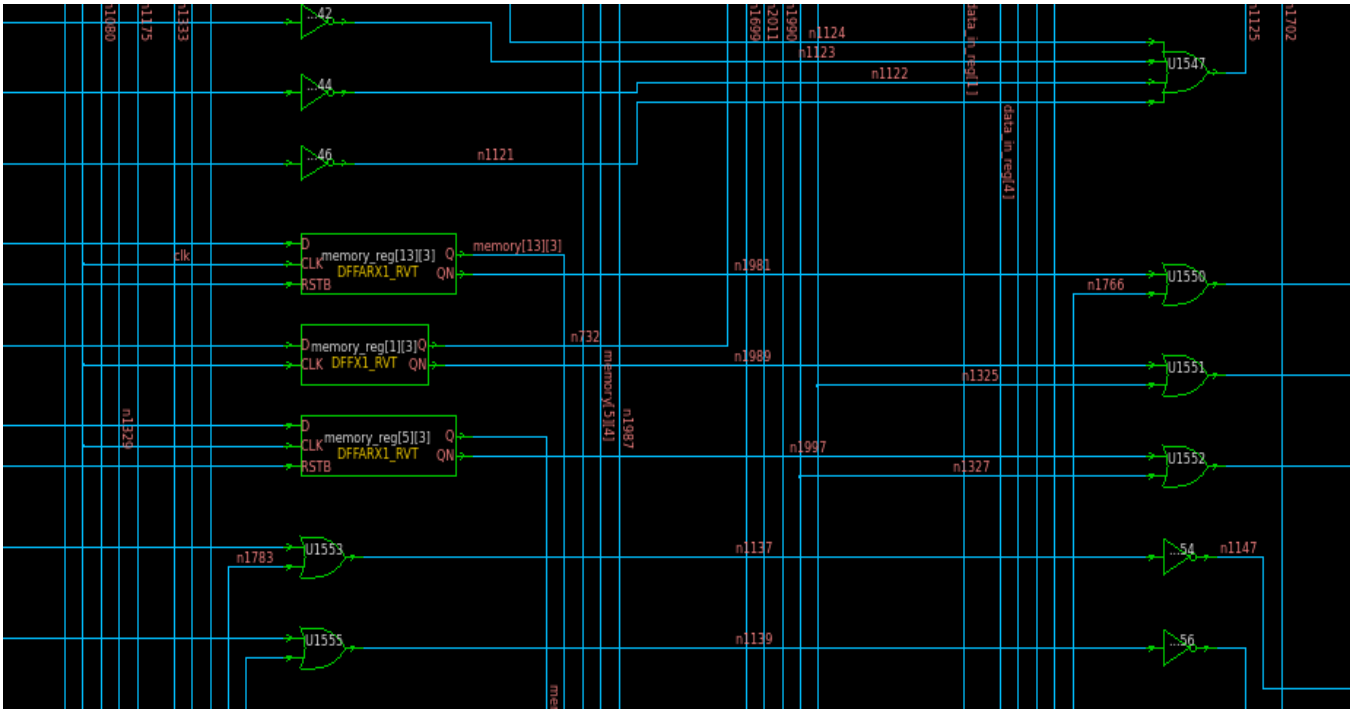
Synthesis by using the SYNOPSYS TOOLS (SDC)-



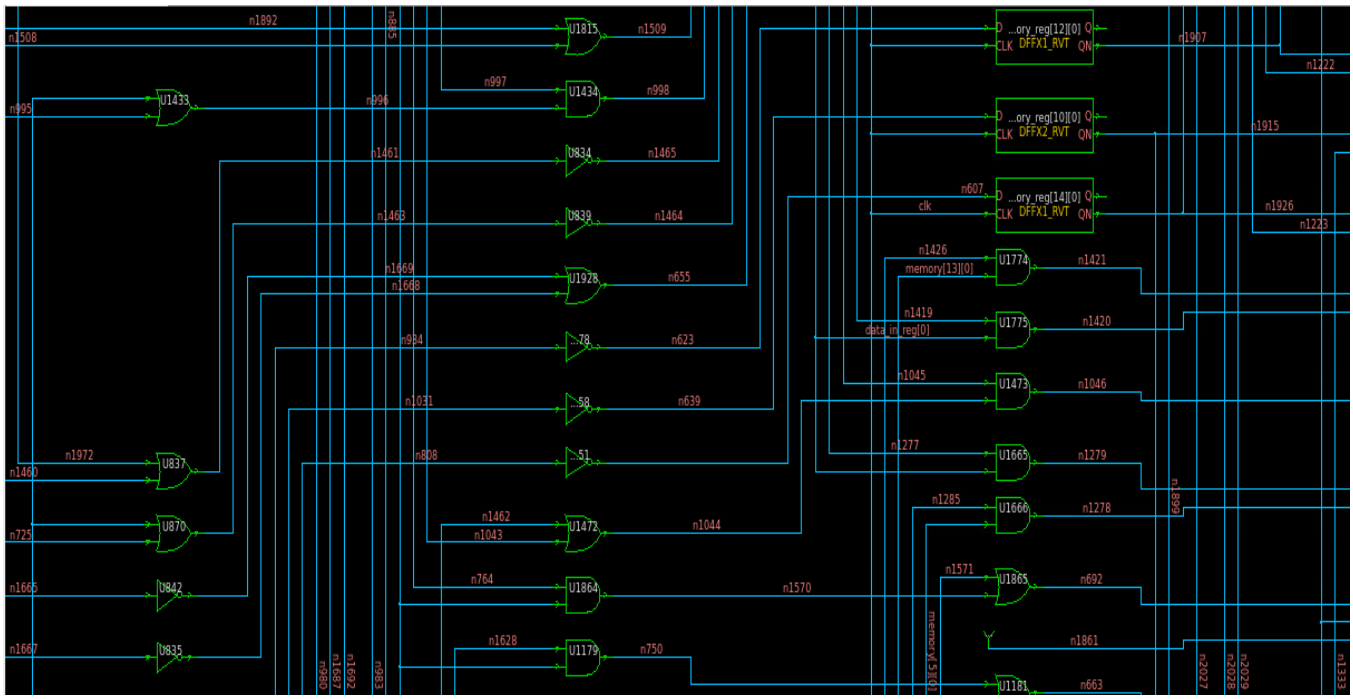
(Schematic View of 16x8sync_fifo - by using Synopsys Design Compiler (SDC))



(This is the internal schematic view of synthesizable circuits)



(Zoom view inside the ckt -By using the Synopsys Design Compiler (SDC))



Simulation Results

Report design Area

Report_timing

dc_shell> report_timing -to data_in[6] -significant_digits 6

Report : timing

-path full

-delay max

-max_paths 1

Design : Sync_fifo_16x8

Version: V-2023.12-SP4

Date : Sat Aug 2 13:42:16 2025

****

Operating Conditions: ff1p16v125c Library: saed32rvt_ff1p16v125c

Wire Load Model Mode: enclosed

Startpoint: data_in[6] (input port clocked by clk)

Endpoint: data_in_reg_reg[6]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Des/Clust/Port	Wire Load Model	Library
Sync_fifo_16x8	8000	saed32rvt_ff1p16v125c

Point	Incr	Path
clock clk (rise edge)	0.000000	0.000000
clock network delay (ideal)	0.470000	0.470000
input external delay	1.000000	1.470000 f

data_in[6] (in) 0.000000 1.470000 f

data_in_reg_reg[6]/D (DFFARX2_RVT) 0.010937 1.480937 f

data arrival time 1.480937

clock clk (rise edge) 2.000000 2.000000

clock network delay (ideal) 0.470000 2.470000

clock uncertainty -0.050000 2.420000

data_in_reg_reg[6]/CLK (DFFARX2_RVT) 0.000000 2.420000 r

library setup time -0.230027 2.189973

data required time 2.189973

---

data required time 2.189973

data arrival time -1.480937

---

slack (MET) 0.709036

1

Finally, we obtained the **slack value** is **greater than 0 .**

Hold report-

dc_shell> report_timing -significant_digits 6 -delay_type min

****

Report : timing

-path full

-delay min

-max_paths 1

Design : Sync_fifo_16x8

Version: V-2023.12-SP4

Date : Sat Aug 2 13:46:54 2025

****

Operating Conditions: ff1p16v125c Library: saed32rvt_ff1p16v125c

Wire Load Model Mode: enclosed

Startpoint: data_out_reg_reg[7]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: data_out_reg[7]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Des/Clust/Port	Wire Load Model	Library
Sync_fifo_16x8	8000	saed32rvt_ff1p16v125c

Point	Incr	Path
clock clk (rise edge)	0.000000	0.000000
clock network delay (ideal)	0.000000	0.000000
data_out_reg_reg[7]/CLK (DFFARX1_RVT)	0.000000	0.000000 r
data_out_reg_reg[7]/Q (DFFARX1_RVT)	0.059911	0.059911 r
data_out_reg[7]/D (DFFASRX1_RVT)	0.038314	0.098225 r
data arrival time		0.098225

clock clk (rise edge) 0.000000 0.000000

clock network delay (ideal) 0.000000 0.000000

data_out_reg[7]/CLK (DFFASRX1_RVT) 0.000000 0.000000 r

library hold time 0.000928 0.000928

data required time 0.000928

---

data required time 0.000928

data arrival time -0.098225

---

slack (MET) 0.097297

1

Report_qor

dc_shell> report_qor

****

Report : qor

Design : Sync_fifo_16x8

Version: V-2023.12-SP4

Date : Sat Aug 2 13:48:18 2025

****

Timing Path Group 'clk'

Levels of Logic: 10.00

Critical Path Length: 1.10

Critical Path Slack: -0.16

Critical Path Clk Period: 2.00

Total Negative Slack: -11.64

No. of Violating Paths: 122.00

Worst Hold Violation: 0.00

Total Hold Violation: 0.00

No. of Hold Violations: 0.00

Cell Count

Hierarchical Cell Count: 0

Hierarchical Port Count: 0

Leaf Cell Count: 1076

Buf/Inv Cell Count: 198

Buf Cell Count: 45

Inv Cell Count: 153

CT Buf/Inv Cell Count: 0

Combinational Cell Count: 907

Sequential Cell Count: 169

Macro Count: 0

Area

Combinational Area: 1781.295369

Noncombinational Area: 1173.637018

Buf/Inv Area: 287.436869

Total Buffer Area: 91.49

Total Inverter Area: 195.95

Macro/Black Box Area: 0.000000

Net Area: 471.019285

Cell Area: 2954.932388

Design Area: 3425.951673

Design Rules

Total Number of Nets: 1165

Nets With Violations: 0

Max Trans Violations: 0

Max Cap Violations: 0

Hostname: prac3.trg.vlsiexpert.in

Compile CPU Statistics

Resource Sharing: 0.00

Logic Optimization: 0.75

Mapping Optimization: 7.72

Overall Compile Time: 12.06

Overall Compile Wall Clock Time: 7.35

Design WNS: 0.16 TNS: 11.64 Number of Violating Paths: 122

Design (Hold) WNS: 0.00 TNS: 0.00 Number of Violating Paths: 0

Compiler_ultra

dc_shell> compile_ultra

RM-Info: Completed script /ve/po_home/b23n_sarif_727/internship_mux/DC/rm_setup/dc_setup.tcl

Loading db file '/ve/po_home/b23n_sarif_727/internship_mux/ref/lib/stdcell_rvt/saed32rvt_ff1p16v125c.db'

Loading db file '/ve/po_home/b23n_sarif_727/internship_mux/ref/lib/stdcell_rvt/saed32rvt_tt0p78vn40c.db'

Loading db file '/ve/po_home/b23n_sarif_727/internship_mux/ref/lib/stdcell_rvt/saed32rvt_ss0p7vn40c.db'

Loading db file '/usr/synopsys/syn/V-2023.12-SP4/libraries/syn/dw_foundation.sldb'

Running PRESTO HDLC

Compiling source file ./../simulation/sync_fifo.sv

Presto compilation completed successfully.

Loading db file '/usr/synopsys/syn/V-2023.12-SP4/libraries/syn/gtech.db'

Loading db file '/usr/synopsys/syn/V-2023.12-SP4/libraries/syn/standard.sldb'

Loading link library '**gtech**'

Running PRESTO HDLC

Warning: ./../simulation/sync_fifo.sv:45: signed to unsigned conversion occurs. (VER-318)

Warning: ./../simulation/sync_fifo.sv:50: signed to unsigned conversion occurs. (VER-318)

Statistics for case statements in always block at line 35 in file

'./../simulation/sync_fifo.sv'

| Line | full/ parallel |

| 53 | auto/auto |

==========

Inferred memory devices in process

in routine Sync_fifo_16x8 line 22 in file

'./../simulation/sync_fifo.sv'.

==========

| Register Name | Type | Width | Bus | MB | AR | AS | SR | SS | ST |

==========

| rd_en_reg_reg | Flip-flop | 1 | N | N | Y | N | N | N | N |

| data_in_reg_reg | Flip-flop | 8 | Y | N | Y | N | N | N | N |

| wr_en_reg_reg | Flip-flop | 1 | N | N | Y | N | N | N | N |

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Inferred memory devices in process

in routine Sync_fifo_16x8 line 35 in file

'./../simulation/sync_fifo.sv'.

==========

| Register Name | Type | Width | Bus | MB | AR | AS | SR | SS | ST |

==========

| memory_reg | Flip-flop | 128 | Y | N | N | N | N | N | N |

| wr_ptr_reg_reg | Flip-flop | 4 | Y | N | Y | N | N | N | N |

| rd_ptr_reg_reg | Flip-flop | 4 | Y | N | Y | N | N | N | N |

| count_reg_reg | Flip-flop | 5 | Y | N | Y | N | N | N | N |

| data_out_reg_reg | Flip-flop | 8 | Y | N | Y | N | N | N |

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Inferred memory devices in process

in routine Sync_fifo_16x8 line 68 in file

'./../simulation/sync_fifo.sv'.

==========

| Register Name | Type | Width | Bus | MB | AR | AS | SR | SS | ST |

==========

| underflow_reg | Flip-flop | 1 | N | N | Y | N | N | N | N |

| overflow_reg | Flip-flop | 1 | N | N | Y | N | N | N | N |

==========

Inferred memory devices in process

in routine Sync_fifo_16x8 line 79 in file

'./../simulation/sync_fifo.sv'.

==========

| Register Name | Type | Width | Bus | MB | AR | AS | SR | SS | ST |

==========

| data_out_reg | Flip-flop | 8 | Y | N | Y | N | N | N | N |

==========

Statistics for MUX_OPs

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| block name/line | Inputs | Outputs | # sel inputs |

==========

| Sync_fifo_16x8/49 | 16 | 8 | 4 |

==========

Presto compilation completed successfully. (Sync_fifo_16x8)

Elaborated 1 design.

Current design is now 'Sync_fifo_16x8'.

Current design is 'Sync_fifo_16x8'.

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| DesignWare Building Block Library | Version | Available |

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| Basic DW Building Blocks | V-2023.12-DWBB_202312.4 | * |

| Licensed DW Building Blocks | V-2023.12-DWBB_202312.4 | * |

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| Flow Information |

| Flow | Design Compiler WLM |

| Command Line | compile_ultra |

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| Design Information | Value |

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Number of Scenarios	0
Leaf Cell Count	284
Number of User Hierarchies	0
Sequential Cell Count	169
Macro Count	0
Number of Power Domains	0
Number of Path Groups	2
Number of VT Class	1
Number of Clocks	1
Number of Dont Touch Cells	45
Number of Dont Touch Nets	0
Number of Size Only Cells	0
Design with UPF Data	false

sync_fifo16x8_bit.sdc

```
create_clock -period 2.0 [get_ports clk]
set_clock_uncertainty -setup 0.05 [get_clocks clk]
set_clock_latency -source -max 0.0700 [get_clocks clk]
set_clock_latency -max 0.400 [get_clocks clk]
set_clock_transition -max 0.25 [get_clocks clk]

set_input_delay -max 1 -clock clk [all_inputs]
set_input_transition -max 1 [all_inputs]

set_output_delay -max 1 -clock clk [all_outputs]
set_load -max 1 [all_outputs]
```

Mapping netlist file (Sync_fifo_16x8.mapped.v) created by tool-

```
////////////////////////////////////////////////////////////
// Created by: Synopsys DC Ultra(TM) in wire load mode
// Version   : V-2023.12-SP4
// Date      : Sat Aug  2 13:59:03 2025
////////////////////////////////////////////////////////////

module Sync_fifo_16x8 ( data_out, underflow, overflow, full, empty, data_in,
        clk, rst_n, rd_en, wr_en );
  output [7:0] data_out;
  input [7:0] data_in;
  input clk, rst_n, rd_en, wr_en;
  output underflow, overflow, full, empty;
  wire   wr_en_reg, rd_en_reg, \memory[15][7] , \memory[15][6] ,
        \memory[15][5] , \memory[15][4] , \memory[15][3] , \memory[15][2] ,
        \memory[15][1] , \memory[15][0] , \memory[14][7] , \memory[14][6] ,
        \memory[14][5] , \memory[14][4] , \memory[14][3] , \memory[14][2] ,
        \memory[14][1] , \memory[14][0] , \memory[13][7] , \memory[13][6] ,
        \memory[13][5] , \memory[13][4] , \memory[13][3] , \memory[13][2] ,
        \memory[13][1] , \memory[13][0] , \memory[12][7] , \memory[12][6] ,
        \memory[12][5] , \memory[12][4] , \memory[12][3] , \memory[12][2] ,
        \memory[12][1] , \memory[12][0] , \memory[11][7] , \memory[11][6] ,
        \memory[11][5] , \memory[11][4] , \memory[11][3] , \memory[11][2] ,
        \memory[11][1] , \memory[11][0] , \memory[10][7] , \memory[10][6] ,
        \memory[10][5] , \memory[10][4] , \memory[10][3] , \memory[10][2] ,
        \memory[10][1] , \memory[10][0] , \memory[9][7] , \memory[9][6] ,
        \memory[9][5] , \memory[9][4] , \memory[9][3] , \memory[9][2] ,
```

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    n1437, n1438, n1439, n1440, n1441, n1442, n1443, n1444, n1445, n1446,
    n1447, n1448, n1449, n1450, n1451, n1452, n1453, n1454, n1455, n1456,
    n1457, n1458, n1459, n1460, n1461, n1462, n1463, n1466, n1467, n1468,
    n1469, n1470, n1471, n1472, n1473, n1474, n1475, n1476, n1477, n1478,
    n1479, n1480, n1481, n1482, n1483, n1485, n1486, n1487, n1488, n1489,
    n1490, n1491, n1492, n1493;
wire   [7:0] data_in_reg;
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  wire   [3:0] wr_ptr_reg;
  wire   [3:0] rd_ptr_reg;
  wire   [4:0] count_reg;
  wire   [7:0] data_out_reg;

  DFFARX1_RVT \data_in_reg_reg[6]  ( .D(data_in[6]), .CLK(clk), .RSTB(n1490),
       .Q(data_in_reg[6]), .QN(n1485) );
  DFFARX1_RVT wr_en_reg_reg ( .D(wr_en), .CLK(clk), .RSTB(n1489), .Q(wr_en_reg)
);
  DFFARX1_RVT underflow_reg ( .D(N129), .CLK(clk), .RSTB(n1489), .Q(underflow)
       );
  DFFARX1_RVT overflow_reg ( .D(N128), .CLK(clk), .RSTB(n1491), .Q(overflow)
       );
  DFFARX1_RVT \data_out_reg[7]_inst  ( .D(data_out_reg[7]), .CLK(clk), .RSTB(
       n1487), .Q(data_out[7]) );
  DFFARX1_RVT \data_out_reg[4]_inst  ( .D(data_out_reg[4]), .CLK(clk), .RSTB(
       n1487), .Q(data_out[4]) );
  DFFARX1_RVT \data_out_reg[3]_inst  ( .D(data_out_reg[3]), .CLK(clk), .RSTB(
       n1489), .Q(data_out[3]) );
  DFFARX1_RVT \wr_ptr_reg_reg[2]  ( .D(n586), .CLK(clk), .RSTB(n1490), .Q(
       wr_ptr_reg[2]), .QN(n724) );
  DFFARX1_RVT \data_out_reg_reg[5]  ( .D(n578), .CLK(clk), .RSTB(n1488), .Q(
       data_out_reg[5]) );
  DFFARX1_RVT \data_out_reg_reg[2]  ( .D(n575), .CLK(clk), .RSTB(n1489), .Q(
       data_out_reg[2]) );
  DFFX1_RVT \memory_reg[0][1]  ( .D(n719), .CLK(clk), .Q(\memory[0][1] ) );
  DFFX1_RVT \memory_reg[0][2]  ( .D(n718), .CLK(clk), .Q(\memory[0][2] ) );
  DFFX1_RVT \memory_reg[2][2]  ( .D(n701), .CLK(clk), .Q(\memory[2][2] ) );
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DFFX1_RVT \memory_reg[2][3]  ( .D(n700), .CLK(clk), .Q(\memory[2][3] ) );
DFFX1_RVT \memory_reg[2][5]  ( .D(n698), .CLK(clk), .Q(\memory[2][5] ) );
DFFX1_RVT \memory_reg[2][7]  ( .D(n696), .CLK(clk), .Q(\memory[2][7] ) );
DFFX1_RVT \memory_reg[3][5]  ( .D(n695), .CLK(clk), .Q(\memory[3][5] ) );
DFFX1_RVT \memory_reg[3][6]  ( .D(n694), .CLK(clk), .Q(\memory[3][6] ) );
DFFX1_RVT \memory_reg[3][0]  ( .D(n692), .CLK(clk), .Q(\memory[3][0] ) );
DFFX1_RVT \memory_reg[3][1]  ( .D(n691), .CLK(clk), .Q(\memory[3][1] ) );
DFFX1_RVT \memory_reg[3][2]  ( .D(n690), .CLK(clk), .Q(\memory[3][2] ) );
DFFX1_RVT \memory_reg[3][3]  ( .D(n689), .CLK(clk), .Q(\memory[3][3] ) );
DFFX1_RVT \memory_reg[3][4]  ( .D(n688), .CLK(clk), .Q(\memory[3][4] ) );
DFFX1_RVT \memory_reg[5][1]  ( .D(n678), .CLK(clk), .Q(\memory[5][1] ) );
DFFX1_RVT \memory_reg[5][3]  ( .D(n676), .CLK(clk), .Q(\memory[5][3] ) );
DFFX1_RVT \memory_reg[5][4]  ( .D(n675), .CLK(clk), .Q(\memory[5][4] ) );
DFFX1_RVT \memory_reg[5][6]  ( .D(n673), .CLK(clk), .Q(\memory[5][6] ) );
DFFX1_RVT \memory_reg[5][7]  ( .D(n672), .CLK(clk), .Q(\memory[5][7] ), .QN(
    n725) );
DFFX1_RVT \memory_reg[7][0]  ( .D(n663), .CLK(clk), .Q(\memory[7][0] ) );
DFFX1_RVT \memory_reg[7][1]  ( .D(n662), .CLK(clk), .Q(\memory[7][1] ) );
DFFX1_RVT \memory_reg[7][2]  ( .D(n661), .CLK(clk), .Q(\memory[7][2] ) );
DFFX1_RVT \memory_reg[7][4]  ( .D(n659), .CLK(clk), .Q(\memory[7][4] ) );
DFFX1_RVT \memory_reg[7][5]  ( .D(n658), .CLK(clk), .Q(\memory[7][5] ) );
DFFX1_RVT \memory_reg[7][6]  ( .D(n657), .CLK(clk), .Q(\memory[7][6] ) );
DFFX1_RVT \memory_reg[8][0]  ( .D(n655), .CLK(clk), .Q(\memory[8][0] ) );
DFFX1_RVT \memory_reg[8][1]  ( .D(n654), .CLK(clk), .Q(\memory[8][1] ) );
DFFX1_RVT \memory_reg[8][2]  ( .D(n653), .CLK(clk), .Q(\memory[8][2] ) );
DFFX1_RVT \memory_reg[8][3]  ( .D(n652), .CLK(clk), .Q(\memory[8][3] ) );
DFFX1_RVT \memory_reg[8][4]  ( .D(n651), .CLK(clk), .Q(\memory[8][4] ) );
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DFFX1_RVT \memory_reg[8][5]  ( .D(n650), .CLK(clk), .Q(\memory[8][5] ) );
DFFX1_RVT \memory_reg[8][6]  ( .D(n649), .CLK(clk), .Q(\memory[8][6] ) );
DFFX1_RVT \memory_reg[8][7]  ( .D(n648), .CLK(clk), .Q(\memory[8][7] ) );
DFFX1_RVT \memory_reg[9][0]  ( .D(n647), .CLK(clk), .Q(\memory[9][0] ) );
DFFX1_RVT \memory_reg[9][1]  ( .D(n646), .CLK(clk), .Q(\memory[9][1] ) );
DFFX1_RVT \memory_reg[9][2]  ( .D(n645), .CLK(clk), .Q(\memory[9][2] ) );
DFFX1_RVT \memory_reg[9][3]  ( .D(n644), .CLK(clk), .Q(\memory[9][3] ) );
DFFX1_RVT \memory_reg[9][4]  ( .D(n643), .CLK(clk), .Q(\memory[9][4] ) );
DFFX1_RVT \memory_reg[9][5]  ( .D(n642), .CLK(clk), .Q(\memory[9][5] ) );
DFFX1_RVT \memory_reg[9][6]  ( .D(n641), .CLK(clk), .Q(\memory[9][6] ) );
DFFX1_RVT \memory_reg[9][7]  ( .D(n640), .CLK(clk), .Q(\memory[9][7] ) );
DFFX1_RVT \memory_reg[10][0]  ( .D(n639), .CLK(clk), .Q(\memory[10][0] ) );
DFFX1_RVT \memory_reg[10][1]  ( .D(n638), .CLK(clk), .Q(\memory[10][1] ) );
DFFX1_RVT \memory_reg[10][2]  ( .D(n637), .CLK(clk), .Q(\memory[10][2] ) );
DFFX1_RVT \memory_reg[10][3]  ( .D(n636), .CLK(clk), .Q(\memory[10][3] ) );
DFFX1_RVT \memory_reg[10][4]  ( .D(n635), .CLK(clk), .Q(\memory[10][4] ) );
DFFX1_RVT \memory_reg[11][0]  ( .D(n631), .CLK(clk), .Q(\memory[11][0] ) );
DFFX1_RVT \memory_reg[11][1]  ( .D(n630), .CLK(clk), .Q(\memory[11][1] ) );
DFFX1_RVT \memory_reg[11][2]  ( .D(n629), .CLK(clk), .Q(\memory[11][2] ) );
DFFX1_RVT \memory_reg[11][4]  ( .D(n627), .CLK(clk), .Q(\memory[11][4] ) );
DFFX1_RVT \memory_reg[11][5]  ( .D(n626), .CLK(clk), .Q(\memory[11][5] ) );
DFFX1_RVT \memory_reg[11][7]  ( .D(n624), .CLK(clk), .Q(\memory[11][7] ) );
DFFX1_RVT \memory_reg[12][0]  ( .D(n623), .CLK(clk), .Q(\memory[12][0] ) );
DFFX1_RVT \memory_reg[12][1]  ( .D(n622), .CLK(clk), .Q(\memory[12][1] ) );
DFFX1_RVT \memory_reg[12][2]  ( .D(n621), .CLK(clk), .Q(\memory[12][2] ) );
DFFX1_RVT \memory_reg[12][4]  ( .D(n619), .CLK(clk), .Q(\memory[12][4] ) );
DFFX1_RVT \memory_reg[12][7]  ( .D(n616), .CLK(clk), .Q(\memory[12][7] ) );
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DFFX1_RVT \memory_reg[14][0]  ( .D(n607), .CLK(clk), .Q(\memory[14][0] ) );
DFFX1_RVT \memory_reg[14][1]  ( .D(n606), .CLK(clk), .Q(\memory[14][1] ) );
DFFX1_RVT \memory_reg[14][2]  ( .D(n605), .CLK(clk), .Q(\memory[14][2] ) );
DFFX1_RVT \memory_reg[14][4]  ( .D(n603), .CLK(clk), .Q(\memory[14][4] ) );
DFFX1_RVT \memory_reg[14][7]  ( .D(n600), .CLK(clk), .Q(\memory[14][7] ) );
DFFX1_RVT \memory_reg[15][0]  ( .D(n599), .CLK(clk), .Q(\memory[15][0] ) );
DFFX1_RVT \memory_reg[15][1]  ( .D(n598), .CLK(clk), .Q(\memory[15][1] ) );
DFFX1_RVT \memory_reg[15][2]  ( .D(n597), .CLK(clk), .Q(\memory[15][2] ) );
DFFX1_RVT \memory_reg[15][3]  ( .D(n596), .CLK(clk), .Q(\memory[15][3] ) );
DFFX1_RVT \memory_reg[15][5]  ( .D(n594), .CLK(clk), .Q(\memory[15][5] ) );
DFFX1_RVT \memory_reg[15][6]  ( .D(n593), .CLK(clk), .Q(\memory[15][6] ) );
DFFX1_RVT \memory_reg[15][7]  ( .D(n592), .CLK(clk), .Q(\memory[15][7] ) );
DFFARX2_RVT \data_in_reg_reg[7]  ( .D(data_in[7]), .CLK(clk), .RSTB(n1489),
      .Q(data_in_reg[7]), .QN(n1486) );
DFFARX1_RVT \count_reg_reg[0]  ( .D(n721), .CLK(clk), .RSTB(n1489), .Q(
      count_reg[0]), .QN(n1466) );
DFFARX1_RVT \wr_ptr_reg_reg[0]  ( .D(n588), .CLK(clk), .RSTB(n1487), .Q(
      wr_ptr_reg[0]), .QN(n1471) );
DFFARX1_RVT \data_out_reg_reg[1]  ( .D(n574), .CLK(clk), .RSTB(n1491), .Q(
      data_out_reg[1]) );
DFFARX1_RVT \rd_ptr_reg_reg[1]  ( .D(n583), .CLK(clk), .RSTB(n1489), .Q(
      rd_ptr_reg[1]), .QN(n1493) );
DFFARX1_RVT \data_out_reg_reg[7]  ( .D(n580), .CLK(clk), .RSTB(n1488), .Q(
      data_out_reg[7]) );
DFFARX1_RVT \data_out_reg_reg[4]  ( .D(n577), .CLK(clk), .RSTB(n1489), .Q(
      data_out_reg[4]) );
DFFARX1_RVT \data_out_reg_reg[0]  ( .D(n573), .CLK(clk), .RSTB(n1487), .Q(
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    data_out_reg[0]) );
DFFARX1_RVT \data_out_reg_reg[6]  ( .D(n579), .CLK(clk), .RSTB(n1491), .Q(
    data_out_reg[6]) );
DFFARX1_RVT \data_out_reg_reg[3]  ( .D(n576), .CLK(clk), .RSTB(n1491), .Q(
    data_out_reg[3]) );
DFFX1_RVT \memory_reg[13][7]  ( .D(n608), .CLK(clk), .Q(\memory[13][7] ) );
DFFX1_RVT \memory_reg[13][6]  ( .D(n609), .CLK(clk), .Q(\memory[13][6] ) );
DFFX1_RVT \memory_reg[13][5]  ( .D(n610), .CLK(clk), .Q(\memory[13][5] ) );
DFFX1_RVT \memory_reg[13][4]  ( .D(n611), .CLK(clk), .Q(\memory[13][4] ) );
DFFX1_RVT \memory_reg[13][3]  ( .D(n612), .CLK(clk), .Q(\memory[13][3] ) );
DFFX1_RVT \memory_reg[13][2]  ( .D(n613), .CLK(clk), .Q(\memory[13][2] ) );
DFFX1_RVT \memory_reg[13][1]  ( .D(n614), .CLK(clk), .Q(\memory[13][1] ) );
DFFX1_RVT \memory_reg[13][0]  ( .D(n615), .CLK(clk), .Q(\memory[13][0] ) );
DFFX1_RVT \memory_reg[2][0]  ( .D(n703), .CLK(clk), .Q(\memory[2][0] ) );
DFFX1_RVT \memory_reg[15][4]  ( .D(n595), .CLK(clk), .Q(\memory[15][4] ) );
DFFX1_RVT \memory_reg[12][5]  ( .D(n618), .CLK(clk), .Q(\memory[12][5] ) );
DFFX1_RVT \memory_reg[14][5]  ( .D(n602), .CLK(clk), .Q(\memory[14][5] ) );
DFFX1_RVT \memory_reg[1][7]  ( .D(n704), .CLK(clk), .Q(\memory[1][7] ) );
DFFX1_RVT \memory_reg[1][6]  ( .D(n705), .CLK(clk), .Q(\memory[1][6] ) );
DFFX1_RVT \memory_reg[1][4]  ( .D(n707), .CLK(clk), .Q(\memory[1][4] ) );
DFFX1_RVT \memory_reg[1][3]  ( .D(n708), .CLK(clk), .Q(\memory[1][3] ) );
DFFX1_RVT \memory_reg[2][6]  ( .D(n697), .CLK(clk), .Q(\memory[2][6] ) );
DFFX1_RVT \memory_reg[11][6]  ( .D(n625), .CLK(clk), .Q(\memory[11][6] ) );
DFFX1_RVT \memory_reg[11][3]  ( .D(n628), .CLK(clk), .Q(\memory[11][3] ) );
DFFX1_RVT \memory_reg[6][3]  ( .D(n668), .CLK(clk), .Q(\memory[6][3] ) );
DFFX1_RVT \memory_reg[5][0]  ( .D(n679), .CLK(clk), .Q(\memory[5][0] ) );
DFFX1_RVT \memory_reg[12][6]  ( .D(n617), .CLK(clk), .Q(\memory[12][6] ) );
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DFFX1_RVT \memory_reg[12][3]  ( .D(n620), .CLK(clk), .Q(\memory[12][3] ) );
DFFX1_RVT \memory_reg[14][6]  ( .D(n601), .CLK(clk), .Q(\memory[14][6] ) );
DFFX1_RVT \memory_reg[14][3]  ( .D(n604), .CLK(clk), .Q(\memory[14][3] ) );
DFFX1_RVT \memory_reg[1][2]  ( .D(n709), .CLK(clk), .Q(\memory[1][2] ) );
DFFX1_RVT \memory_reg[0][4]  ( .D(n716), .CLK(clk), .Q(\memory[0][4] ) );
DFFX1_RVT \memory_reg[0][5]  ( .D(n715), .CLK(clk), .Q(\memory[0][5] ) );
DFFX1_RVT \memory_reg[0][7]  ( .D(n713), .CLK(clk), .Q(\memory[0][7] ) );
DFFX1_RVT \memory_reg[0][6]  ( .D(n714), .CLK(clk), .Q(\memory[0][6] ) );
DFFX1_RVT \memory_reg[6][5]  ( .D(n666), .CLK(clk), .Q(\memory[6][5] ) );
DFFX1_RVT \memory_reg[5][5]  ( .D(n674), .CLK(clk), .Q(\memory[5][5] ) );
DFFX1_RVT \memory_reg[6][6]  ( .D(n665), .CLK(clk), .Q(\memory[6][6] ) );
DFFX1_RVT \memory_reg[6][4]  ( .D(n667), .CLK(clk), .Q(\memory[6][4] ) );
DFFX1_RVT \memory_reg[6][0]  ( .D(n671), .CLK(clk), .Q(\memory[6][0] ) );
DFFX1_RVT \memory_reg[5][2]  ( .D(n677), .CLK(clk), .Q(\memory[5][2] ) );
DFFX1_RVT \memory_reg[6][7]  ( .D(n664), .CLK(clk), .Q(\memory[6][7] ) );
DFFX1_RVT \memory_reg[3][7]  ( .D(n693), .CLK(clk), .Q(\memory[3][7] ) );
DFFX1_RVT \memory_reg[1][5]  ( .D(n706), .CLK(clk), .Q(\memory[1][5] ) );
DFFX1_RVT \memory_reg[1][1]  ( .D(n710), .CLK(clk), .Q(\memory[1][1] ) );
DFFX1_RVT \memory_reg[1][0]  ( .D(n711), .CLK(clk), .Q(\memory[1][0] ) );
DFFX1_RVT \memory_reg[7][3]  ( .D(n660), .CLK(clk), .Q(\memory[7][3] ) );
DFFX1_RVT \memory_reg[0][0]  ( .D(n712), .CLK(clk), .Q(\memory[0][0] ) );
DFFX1_RVT \memory_reg[6][2]  ( .D(n669), .CLK(clk), .Q(\memory[6][2] ) );
DFFX1_RVT \memory_reg[6][1]  ( .D(n670), .CLK(clk), .Q(\memory[6][1] ) );
DFFX1_RVT \memory_reg[2][4]  ( .D(n699), .CLK(clk), .Q(\memory[2][4] ) );
DFFX1_RVT \memory_reg[2][1]  ( .D(n702), .CLK(clk), .Q(\memory[2][1] ) );
DFFARX1_RVT \memory_reg[7][7]  ( .D(n656), .CLK(clk), .RSTB(1'b1), .Q(
    \memory[7][7] ), .QN(n1481) );
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DFFARX1_RVT \memory_reg[0][3]  ( .D(n717), .CLK(clk), .RSTB(1'b1), .Q(
      \memory[0][3] ) );
DFFARX1_RVT \count_reg_reg[3]  ( .D(n589), .CLK(clk), .RSTB(n1487), .Q(
      count_reg[3]), .QN(n1478) );
DFFX2_RVT \memory_reg[4][2]  ( .D(n685), .CLK(clk), .Q(\memory[4][2] ) );
DFFX2_RVT \memory_reg[4][0]  ( .D(n687), .CLK(clk), .Q(\memory[4][0] ) );
DFFX2_RVT \memory_reg[4][5]  ( .D(n682), .CLK(clk), .Q(\memory[4][5] ) );
DFFX2_RVT \memory_reg[4][3]  ( .D(n684), .CLK(clk), .Q(\memory[4][3] ) );
DFFX2_RVT \memory_reg[4][7]  ( .D(n680), .CLK(clk), .Q(\memory[4][7] ) );
DFFX2_RVT \memory_reg[4][6]  ( .D(n681), .CLK(clk), .Q(\memory[4][6] ) );
DFFX2_RVT \memory_reg[4][4]  ( .D(n683), .CLK(clk), .Q(\memory[4][4] ) );
DFFARX1_RVT \count_reg_reg[4]  ( .D(n720), .CLK(clk), .RSTB(n1488), .Q(
      count_reg[4]), .QN(n1480) );
DFFARX1_RVT \count_reg_reg[1]  ( .D(n591), .CLK(clk), .RSTB(n1491), .Q(
      count_reg[1]), .QN(n1470) );
DFFARX1_RVT \count_reg_reg[2]  ( .D(n590), .CLK(clk), .RSTB(n1488), .Q(
      count_reg[2]), .QN(n1467) );
DFFARX1_RVT \rd_ptr_reg_reg[2]  ( .D(n582), .CLK(clk), .RSTB(n1489), .Q(
      rd_ptr_reg[2]), .QN(n1472) );
DFFARX1_RVT \rd_ptr_reg_reg[3]  ( .D(n581), .CLK(clk), .RSTB(n1489), .Q(
      rd_ptr_reg[3]), .QN(n1468) );
DFFARX1_RVT \wr_ptr_reg_reg[1]  ( .D(n587), .CLK(clk), .RSTB(n1488), .Q(
      wr_ptr_reg[1]), .QN(n1479) );
DFFARX1_RVT \wr_ptr_reg_reg[3]  ( .D(n585), .CLK(clk), .RSTB(n1491), .Q(
      wr_ptr_reg[3]), .QN(n1482) );
DFFX1_RVT \memory_reg[10][5]  ( .D(n634), .CLK(clk), .Q(\memory[10][5] ) );
DFFX1_RVT \memory_reg[10][7]  ( .D(n632), .CLK(clk), .Q(\memory[10][7] ) );
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  DFFX1_RVT \memory_reg[10][6]  ( .D(n633), .CLK(clk), .Q(\memory[10][6] ) );
  DFFARX1_RVT rd_en_reg_reg ( .D(rd_en), .CLK(clk), .RSTB(n1489), .Q(rd_en_reg)
);
  DFFARX1_RVT \rd_ptr_reg_reg[0]  ( .D(n584), .CLK(clk), .RSTB(n1487), .Q(
        n1492), .QN(n1469) );
  DFFX1_RVT \memory_reg[4][1]  ( .D(n686), .CLK(clk), .Q(\memory[4][1] ) );
  DFFARX1_RVT \data_in_reg_reg[5]  ( .D(data_in[5]), .CLK(clk), .RSTB(n1491),
        .Q(data_in_reg[5]), .QN(n1483) );
  DFFARX1_RVT \data_in_reg_reg[0]  ( .D(data_in[0]), .CLK(clk), .RSTB(n1489),
        .Q(data_in_reg[0]), .QN(n1475) );
  DFFARX1_RVT \data_in_reg_reg[3]  ( .D(data_in[3]), .CLK(clk), .RSTB(n1487),
        .Q(data_in_reg[3]), .QN(n1473) );
  DFFARX1_RVT \data_in_reg_reg[4]  ( .D(data_in[4]), .CLK(clk), .RSTB(n1488),
        .Q(data_in_reg[4]), .QN(n1474) );
  DFFARX1_RVT \data_in_reg_reg[2]  ( .D(data_in[2]), .CLK(clk), .RSTB(n1488),
        .Q(data_in_reg[2]), .QN(n1476) );
  DFFARX1_RVT \data_in_reg_reg[1]  ( .D(data_in[1]), .CLK(clk), .RSTB(n1489),
        .Q(data_in_reg[1]), .QN(n1477) );
  DFFARX1_RVT \data_out_reg[0]_inst  ( .D(data_out_reg[0]), .CLK(clk), .RSTB(
        n1489), .Q(data_out[0]) );
  DFFARX1_RVT \data_out_reg[6]_inst  ( .D(data_out_reg[6]), .CLK(clk), .RSTB(
        n1489), .Q(data_out[6]) );
  DFFARX1_RVT \data_out_reg[2]_inst  ( .D(data_out_reg[2]), .CLK(clk), .RSTB(
        n1489), .Q(data_out[2]) );
  DFFARX1_RVT \data_out_reg[5]_inst  ( .D(data_out_reg[5]), .CLK(clk), .RSTB(
        n1489), .Q(data_out[5]) );
  DFFARX1_RVT \data_out_reg[1]_inst  ( .D(data_out_reg[1]), .CLK(clk), .RSTB(
        n1489), .Q(data_out[1]) );
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INVX0_RVT U728 ( .A(n792), .Y(n795) );
OR2X1_RVT U729 ( .A1(wr_ptr_reg[0]), .A2(n1479), .Y(n1101) );
OR2X2_RVT U730 ( .A1(n1129), .A2(n1101), .Y(n792) );
OR3X2_RVT U731 ( .A1(n1211), .A2(n1479), .A3(n1471), .Y(n1212) );
OR2X2_RVT U732 ( .A1(n1486), .A2(n1413), .Y(n1394) );
OR2X2_RVT U733 ( .A1(n1485), .A2(n1413), .Y(n812) );
OR2X2_RVT U734 ( .A1(n1483), .A2(n1413), .Y(n805) );
INVX0_RVT U735 ( .A(n1182), .Y(n1231) );
INVX0_RVT U736 ( .A(n1234), .Y(n1243) );
INVX0_RVT U737 ( .A(n1170), .Y(n1177) );
INVX0_RVT U738 ( .A(n1135), .Y(n1173) );
INVX0_RVT U739 ( .A(n1259), .Y(n1263) );
INVX0_RVT U740 ( .A(n1135), .Y(n1176) );
INVX0_RVT U741 ( .A(n935), .Y(n997) );
INVX0_RVT U742 ( .A(n1256), .Y(n1259) );
INVX0_RVT U743 ( .A(n760), .Y(n1208) );
INVX0_RVT U744 ( .A(n792), .Y(n935) );
INVX1_RVT U745 ( .A(n1130), .Y(n1140) );
INVX0_RVT U746 ( .A(n767), .Y(n1205) );
INVX1_RVT U747 ( .A(n1212), .Y(n1256) );
INVX0_RVT U748 ( .A(n1288), .Y(n1292) );
INVX1_RVT U749 ( .A(n1212), .Y(n1221) );
INVX0_RVT U750 ( .A(n931), .Y(n1013) );
INVX1_RVT U751 ( .A(n931), .Y(n1008) );
INVX1_RVT U752 ( .A(n1009), .Y(n931) );
INVX1_RVT U753 ( .A(n1009), .Y(n1020) );
INVX0_RVT U754 ( .A(n1161), .Y(n1167) );
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INVX1_RVT U755 ( .A(n1102), .Y(n1123) );
INVX0_RVT U756 ( .A(n917), .Y(n919) );
INVX1_RVT U757 ( .A(n1112), .Y(n1117) );
INVX0_RVT U758 ( .A(n1368), .Y(n1346) );
INVX0_RVT U759 ( .A(n1475), .Y(n1293) );
INVX0_RVT U760 ( .A(n1475), .Y(n1235) );
INVX0_RVT U761 ( .A(n1485), .Y(n1277) );
INVX0_RVT U762 ( .A(n1474), .Y(n1195) );
INVX0_RVT U763 ( .A(n1473), .Y(n1415) );
INVX0_RVT U764 ( .A(n1477), .Y(n1239) );
INVX0_RVT U765 ( .A(n1483), .Y(n1187) );
INVX0_RVT U766 ( .A(n1476), .Y(n1244) );
INVX0_RVT U767 ( .A(n1469), .Y(n1368) );
INVX0_RVT U768 ( .A(n1486), .Y(n1281) );
INVX0_RVT U769 ( .A(n730), .Y(n729) );
INVX0_RVT U770 ( .A(n941), .Y(n942) );
INVX0_RVT U771 ( .A(n1247), .Y(n1253) );
INVX0_RVT U772 ( .A(n974), .Y(n994) );
INVX0_RVT U773 ( .A(n1202), .Y(n763) );
INVX0_RVT U774 ( .A(n1192), .Y(n1238) );
INVX0_RVT U775 ( .A(n980), .Y(n993) );
INVX0_RVT U776 ( .A(n1143), .Y(n1096) );
INVX0_RVT U777 ( .A(n1256), .Y(n1262) );
INVX0_RVT U778 ( .A(n1225), .Y(n1224) );
INVX0_RVT U779 ( .A(n1200), .Y(n1252) );
INVX0_RVT U780 ( .A(n935), .Y(n940) );
INVX0_RVT U781 ( .A(n1143), .Y(n1093) );
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INVX0_RVT U782 (.A(n801), .Y(n802));
INVX0_RVT U783 (.A(n739), .Y(n1412));
INVX0_RVT U784 (.A(n1410), .Y(n1411));
INVX0_RVT U785 (.A(n1135), .Y(n1170));
INVX0_RVT U786 (.A(n809), .Y(n810));
INVX0_RVT U787 (.A(n980), .Y(n974));
OR2X1_RVT U788 (.A1(n725), .A2(n1405), .Y(n739));
INVX1_RVT U789 (.A(n745), .Y(n1097));
INVX0_RVT U790 (.A(n928), .Y(n929));
INVX0_RVT U791 (.A(n932), .Y(n933));
INVX0_RVT U792 (.A(n745), .Y(n1143));
INVX0_RVT U793 (.A(n727), .Y(n1200));
INVX0_RVT U794 (.A(n727), .Y(n1202));
INVX1_RVT U795 (.A(n1130), .Y(n1135));
INVX1_RVT U796 (.A(n977), .Y(n988));
INVX0_RVT U797 (.A(n977), .Y(n980));
OR2X1_RVT U798 (.A1(n1101), .A2(n766), .Y(n727));
OR3X1_RVT U799 (.A1(n1471), .A2(n1479), .A3(n1129), .Y(n1130));
OR2X1_RVT U800 (.A1(n766), .A2(n972), .Y(n767));
OR2X1_RVT U801 (.A1(n766), .A2(n1268), .Y(n760));
INVX0_RVT U802 (.A(n1274), .Y(n1291));
INVX0_RVT U803 (.A(n922), .Y(n721));
INVX0_RVT U804 (.A(n924), .Y(n926));
INVX0_RVT U805 (.A(n1269), .Y(n1274));
NBUFFX2_RVT U806 (.A(n1420), .Y(n1430));
INVX0_RVT U807 (.A(n1152), .Y(n1158));
INVX0_RVT U808 (.A(n1117), .Y(n1152));

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INVX0_RVT U809 ( .A(n955), .Y(n965) );
INVX1_RVT U810 ( .A(n1117), .Y(n1157) );
INVX0_RVT U811 ( .A(n1429), .Y(n1420) );
INVX1_RVT U812 ( .A(n1122), .Y(n1166) );
INVX0_RVT U813 ( .A(n955), .Y(n968) );
OR2X1_RVT U814 ( .A1(n1452), .A2(n927), .Y(n1009) );
INVX0_RVT U815 ( .A(n945), .Y(n944) );
INVX0_RVT U816 ( .A(n1429), .Y(n1426) );
INVX0_RVT U817 ( .A(n1122), .Y(n1161) );
INVX0_RVT U818 ( .A(n1461), .Y(n1460) );
INVX0_RVT U819 ( .A(n737), .Y(n736) );
INVX1_RVT U820 ( .A(n950), .Y(n955) );
INVX1_RVT U821 ( .A(n950), .Y(n962) );
INVX0_RVT U822 ( .A(n816), .Y(n1423) );
INVX0_RVT U823 ( .A(n1102), .Y(n1122) );
INVX1_RVT U824 ( .A(n1112), .Y(n1126) );
OR3X1_RVT U825 ( .A1(n1482), .A2(n1101), .A3(n1100), .Y(n1102) );
INVX1_RVT U826 ( .A(n1456), .Y(n1324) );
INVX0_RVT U827 ( .A(n1321), .Y(n1353) );
OR2X1_RVT U828 ( .A1(n1437), .A2(n1435), .Y(n917) );
AND3X1_RVT U829 ( .A1(n1482), .A2(n910), .A3(n1450), .Y(n816) );
INVX0_RVT U830 ( .A(n1450), .Y(n1449) );
INVX0_RVT U831 ( .A(n912), .Y(n738) );
INVX1_RVT U832 ( .A(n749), .Y(n1456) );
INVX0_RVT U833 ( .A(n749), .Y(n1321) );
AND2X1_RVT U834 ( .A1(n1444), .A2(n749), .Y(n1437) );
INVX1_RVT U835 ( .A(n821), .Y(n749) );
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INVX1_RVT U836 (.A(n1444), .Y(n912));

INVX0_RVT U837 (.A(n923), .Y(empty));

AND2X1_RVT U838 (.A1(n740), .A2(wr_en_reg), .Y(n1444));

INVX0_RVT U839 (.A(n779), .Y(n740));

INVX0_RVT U840 (.A(n748), .Y(n751));

INVX0_RVT U841 (.A(n1483), .Y(n1431));

OR2X1_RVT U842 (.A1(wr_ptr_reg[1]), .A2(n1471), .Y(n972));

OR2X1_RVT U843 (.A1(wr_ptr_reg[0]), .A2(wr_ptr_reg[1]), .Y(n1268));

INVX0_RVT U844 (.A(n1269), .Y(n1280));

INVX0_RVT U845 (.A(n808), .Y(n1409));

IBUFFX16_RVT U846 (.A(rst_n), .Y(n735));

OR2X1_RVT U847 (.A1(n1267), .A2(n1268), .Y(n1269));

NBUFFX2_RVT U848 (.A(n1292), .Y(n726));

OR2X1_RVT U849 (.A1(n1267), .A2(n1101), .Y(n741));

OR2X1_RVT U850 (.A1(n729), .A2(n1201), .Y(n636));

NBUFFX2_RVT U851 (.A(n1202), .Y(n728));

OR2X1_RVT U852 (.A1(n1473), .A2(n763), .Y(n730));

NBUFFX2_RVT U853 (.A(n741), .Y(n731));

NBUFFX2_RVT U854 (.A(count_reg[4]), .Y(n732));

NBUFFX2_RVT U855 (.A(n1288), .Y(n733));

INVX0_RVT U856 (.A(n816), .Y(n734));

OR2X1_RVT U857 (.A1(n759), .A2(n1482), .Y(n1211));

OR2X1_RVT U858 (.A1(n736), .A2(n735), .Y(n759));

INVX4_RVT U859 (.A(n911), .Y(n1489));

INVX0_RVT U860 (.A(n760), .Y(n1225));

OR2X1_RVT U861 (.A1(n1236), .A2(n1237), .Y(n647));

OR2X1_RVT U862 (.A1(n1245), .A2(n1246), .Y(n645));

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INVX0_RVT U863 ( .A(n767), .Y(n1192) );
OR2X1_RVT U864 ( .A1(n1196), .A2(n1197), .Y(n643) );
INVX0_RVT U865 ( .A(n1394), .Y(n1395) );
INVX0_RVT U866 ( .A(n949), .Y(n1111) );
INVX0_RVT U867 ( .A(n805), .Y(n806) );
INVX0_RVT U868 ( .A(n812), .Y(n813) );
INVX0_RVT U869 ( .A(n742), .Y(n1413) );
INVX0_RVT U870 ( .A(n1274), .Y(n1288) );
OR2X1_RVT U871 ( .A1(n1289), .A2(n1290), .Y(n686) );
OR2X1_RVT U872 ( .A1(n1279), .A2(n1278), .Y(n681) );
OR2X1_RVT U873 ( .A1(n1272), .A2(n1273), .Y(n684) );
OR2X1_RVT U874 ( .A1(n1287), .A2(n1286), .Y(n685) );
OR2X1_RVT U875 ( .A1(n1232), .A2(n1233), .Y(n655) );
INVX0_RVT U876 ( .A(n1225), .Y(n1182) );
AND2X1_RVT U877 ( .A1(n738), .A2(n724), .Y(n737) );
OR2X1_RVT U878 ( .A1(n1183), .A2(n1184), .Y(n651) );
OR2X1_RVT U879 ( .A1(n1190), .A2(n1191), .Y(n650) );
OR2X1_RVT U880 ( .A1(n1399), .A2(n1400), .Y(n678) );
OR2X1_RVT U881 ( .A1(n1406), .A2(n1407), .Y(n675) );
OR2X1_RVT U882 ( .A1(n810), .A2(n811), .Y(n673) );
OR2X1_RVT U883 ( .A1(n1284), .A2(n1285), .Y(n676) );
NBUFFX2_RVT U884 ( .A(n1234), .Y(n1242) );
NBUFFX2_RVT U885 ( .A(n1182), .Y(n1230) );
OR2X1_RVT U886 ( .A1(n1412), .A2(n1411), .Y(n672) );
NBUFFX2_RVT U887 ( .A(n800), .Y(n808) );
INVX0_RVT U888 ( .A(n910), .Y(n911) );
NBUFFX2_RVT U889 ( .A(rst_n), .Y(n815) );
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NBUFFX2_RVT U890 ( .A(n1472), .Y(n828) );
AND3X1_RVT U891 ( .A1(rd_ptr_reg[3]), .A2(n1493), .A3(n828), .Y(n891) );
OR2X1_RVT U892 ( .A1(n1358), .A2(n1357), .Y(n1359) );
INVX0_RVT U893 ( .A(n1376), .Y(n849) );
OR4X1_RVT U894 ( .A1(n868), .A2(n867), .A3(n866), .A4(n865), .Y(n869) );
OR4X1_RVT U895 ( .A1(n841), .A2(n840), .A3(n839), .A4(n838), .Y(n842) );
INVX0_RVT U896 ( .A(n1023), .Y(n1070) );
OR2X1_RVT U897 ( .A1(n1363), .A2(n1362), .Y(n1364) );
NBUFFX2_RVT U898 ( .A(n973), .Y(n791) );
INVX0_RVT U899 ( .A(n785), .Y(n786) );
NBUFFX2_RVT U900 ( .A(n1470), .Y(n1436) );
OR2X1_RVT U901 ( .A1(n790), .A2(n789), .Y(n589) );
OR2X1_RVT U902 ( .A1(n1425), .A2(n1424), .Y(n660) );
OR2X1_RVT U903 ( .A1(n1397), .A2(n1398), .Y(n679) );
INVX0_RVT U904 ( .A(n916), .Y(n583) );
OR2X1_RVT U905 ( .A1(n1433), .A2(n1432), .Y(n658) );
INVX0_RVT U906 ( .A(n911), .Y(n1488) );
AND4X1_RVT U909 ( .A1(n1478), .A2(n1467), .A3(n1470), .A4(n1466), .Y(n748)
    );
AND2X1_RVT U910 ( .A1(count_reg[4]), .A2(n748), .Y(n779) );
AND3X1_RVT U911 ( .A1(n815), .A2(wr_ptr_reg[2]), .A3(n1444), .Y(n949) );
OR2X1_RVT U912 ( .A1(n1111), .A2(wr_ptr_reg[3]), .Y(n1266) );
NBUFFX2_RVT U913 ( .A(n1266), .Y(n798) );
INVX0_RVT U914 ( .A(n741), .Y(n742) );
AND2X1_RVT U915 ( .A1(\memory[6][4] ), .A2(n731), .Y(n744) );
INVX0_RVT U916 ( .A(n742), .Y(n775) );
INVX1_RVT U917 ( .A(n775), .Y(n1414) );
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AND2X1_RVT U918 ( .A1(n1195), .A2(n772), .Y(n743) );
OR2X1_RVT U919 ( .A1(n744), .A2(n743), .Y(n667) );
OR2X1_RVT U920 ( .A1(wr_ptr_reg[3]), .A2(n759), .Y(n973) );
OR2X1_RVT U921 ( .A1(n1268), .A2(n791), .Y(n745) );
AND2X1_RVT U922 ( .A1(data_in_reg[6]), .A2(n1097), .Y(n747) );
AND2X1_RVT U923 ( .A1(\memory[0][6] ), .A2(n1096), .Y(n746) );
OR2X1_RVT U924 ( .A1(n747), .A2(n746), .Y(n714) );
OR2X1_RVT U925 ( .A1(n751), .A2(n732), .Y(n923) );
AND2X1_RVT U926 ( .A1(rd_en_reg), .A2(n923), .Y(n821) );
AND3X1_RVT U927 ( .A1(count_reg[3]), .A2(count_reg[2]), .A3(count_reg[1]),
    .Y(n752) );
AND4X1_RVT U928 ( .A1(count_reg[0]), .A2(n1437), .A3(n752), .A4(n1480), .Y(
    n758) );
NBUFFX2_RVT U929 ( .A(n1466), .Y(n918) );
AND2X1_RVT U930 ( .A1(n1437), .A2(n918), .Y(n750) );
AND2X1_RVT U931 ( .A1(n821), .A2(n912), .Y(n1435) );
OR2X1_RVT U932 ( .A1(n750), .A2(n919), .Y(n1434) );
AND2X1_RVT U933 ( .A1(n1435), .A2(n751), .Y(n755) );
INVX0_RVT U934 ( .A(n752), .Y(n753) );
AND2X1_RVT U935 ( .A1(n1437), .A2(n753), .Y(n754) );
OR3X1_RVT U936 ( .A1(n1434), .A2(n755), .A3(n754), .Y(n756) );
AND2X1_RVT U937 ( .A1(n732), .A2(n756), .Y(n757) );
OR2X1_RVT U938 ( .A1(n758), .A2(n757), .Y(n720) );
NBUFFX2_RVT U939 ( .A(n1211), .Y(n766) );
AND2X1_RVT U940 ( .A1(data_in_reg[7]), .A2(n1208), .Y(n762) );
AND2X1_RVT U941 ( .A1(\memory[8][7] ), .A2(n1182), .Y(n761) );
OR2X1_RVT U942 ( .A1(n762), .A2(n761), .Y(n648) );
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AND2X1_RVT U943 ( .A1(n1281), .A2(n728), .Y(n765) );
AND2X1_RVT U944 ( .A1(\memory[10][7] ), .A2(n1247), .Y(n764) );
OR2X1_RVT U945 ( .A1(n765), .A2(n764), .Y(n632) );
AND2X1_RVT U946 ( .A1(n1281), .A2(n1205), .Y(n769) );
AND2X1_RVT U947 ( .A1(\memory[9][7] ), .A2(n1238), .Y(n768) );
OR2X1_RVT U948 ( .A1(n769), .A2(n768), .Y(n640) );
AND2X1_RVT U949 ( .A1(\memory[6][0] ), .A2(n731), .Y(n771) );
NBUFFX2_RVT U950 ( .A(n1414), .Y(n772) );
AND2X1_RVT U951 ( .A1(n1235), .A2(n772), .Y(n770) );
OR2X1_RVT U952 ( .A1(n771), .A2(n770), .Y(n671) );
AND2X1_RVT U953 ( .A1(\memory[6][1] ), .A2(n731), .Y(n774) );
NBUFFX2_RVT U954 ( .A(n1414), .Y(n776) );
AND2X1_RVT U955 ( .A1(n1239), .A2(n776), .Y(n773) );
OR2X1_RVT U956 ( .A1(n774), .A2(n773), .Y(n670) );
NBUFFX2_RVT U957 ( .A(n775), .Y(n804) );
AND2X1_RVT U958 ( .A1(\memory[6][2] ), .A2(n804), .Y(n778) );
AND2X1_RVT U959 ( .A1(n1244), .A2(n776), .Y(n777) );
OR2X1_RVT U960 ( .A1(n778), .A2(n777), .Y(n669) );
NBUFFX2_RVT U961 ( .A(n779), .Y(full) );
AND2X1_RVT U962 ( .A1(count_reg[2]), .A2(n1435), .Y(n781) );
NBUFFX2_RVT U963 ( .A(n1467), .Y(n946) );
AND2X1_RVT U964 ( .A1(n1437), .A2(n946), .Y(n780) );
AND3X1_RVT U965 ( .A1(count_reg[1]), .A2(count_reg[0]), .A3(n1437), .Y(n783)
    );
AND3X1_RVT U966 ( .A1(n1435), .A2(n1436), .A3(n918), .Y(n1443) );
OR2X1_RVT U967 ( .A1(n783), .A2(n1443), .Y(n945) );
OR3X1_RVT U968 ( .A1(n781), .A2(n780), .A3(n944), .Y(n782) );
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AND2X1_RVT U969 (.A1(n782), .A2(count_reg[3]), .Y(n790));

AND3X1_RVT U970 (.A1(count_reg[2]), .A2(n783), .A3(n1478), .Y(n784));

INVX0_RVT U971 (.A(n784), .Y(n787));

AND2X1_RVT U972 (.A1(rd_en_reg), .A2(full), .Y(n785));

AND2X1_RVT U973 (.A1(n787), .A2(n786), .Y(n788));

INVX0_RVT U974 (.A(n788), .Y(n789));

NBUFFX2_RVT U975 (.A(n973), .Y(n1129));

NBUFFX2_RVT U976 (.A(n795), .Y(n1000));

AND2X1_RVT U977 (.A1(data_in_reg[0]), .A2(n1000), .Y(n794));

AND2X1_RVT U978 (.A1(\memory[2][0]), .A2(n997), .Y(n793));

OR2X1_RVT U979 (.A1(n794), .A2(n793), .Y(n703));

NBUFFX2_RVT U980 (.A(n795), .Y(n1003));

AND2X1_RVT U981 (.A1(data_in_reg[6]), .A2(n1003), .Y(n797));

AND2X1_RVT U982 (.A1(\memory[2][6]), .A2(n997), .Y(n796));

OR2X1_RVT U983 (.A1(n797), .A2(n796), .Y(n697));

OR2X1_RVT U984 (.A1(n972), .A2(n798), .Y(n799));

INVX0_RVT U985 (.A(n799), .Y(n800));

INVX0_RVT U986 (.A(n800), .Y(n1401));

AND2X1_RVT U987 (.A1(\memory[5][5]), .A2(n1401), .Y(n803));

OR2X1_RVT U988 (.A1(n1408), .A2(n1483), .Y(n801));

OR2X1_RVT U989 (.A1(n803), .A2(n802), .Y(n674));

AND2X1_RVT U990 (.A1(\memory[6][5]), .A2(n804), .Y(n807));

OR2X1_RVT U991 (.A1(n807), .A2(n806), .Y(n666));

INVX0_RVT U992 (.A(n808), .Y(n1408));

AND2X1_RVT U993 (.A1(\memory[5][6]), .A2(n1401), .Y(n811));

OR2X1_RVT U994 (.A1(n1485), .A2(n1408), .Y(n809));

AND2X1_RVT U995 (.A1(\memory[6][6]), .A2(n731), .Y(n814));

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OR2X1_RVT U996 ( .A1(n814), .A2(n813), .Y(n665) );
INVX0_RVT U997 ( .A(n815), .Y(n927) );
INVX0_RVT U998 ( .A(n927), .Y(n910) );
AND3X1_RVT U999 ( .A1(n1444), .A2(wr_ptr_reg[0]), .A3(wr_ptr_reg[1]), .Y(
    n1447) );
AND2X1_RVT U1000 ( .A1(n1447), .A2(wr_ptr_reg[2]), .Y(n1450) );
INVX0_RVT U1001 ( .A(n816), .Y(n1429) );
AND2X1_RVT U1002 ( .A1(\memory[7][0] ), .A2(n734), .Y(n818) );
AND2X1_RVT U1003 ( .A1(n1293), .A2(n1420), .Y(n817) );
OR2X1_RVT U1004 ( .A1(n818), .A2(n817), .Y(n663) );
AND2X1_RVT U1005 ( .A1(\memory[7][6] ), .A2(n1423), .Y(n820) );
AND2X1_RVT U1006 ( .A1(n1277), .A2(n1420), .Y(n819) );
OR2X1_RVT U1007 ( .A1(n820), .A2(n819), .Y(n657) );
AND2X1_RVT U1008 ( .A1(data_out_reg[6]), .A2(n1324), .Y(n855) );
INVX0_RVT U1009 ( .A(n1472), .Y(n1458) );
AND3X1_RVT U1010 ( .A1(rd_ptr_reg[3]), .A2(n1458), .A3(n1493), .Y(n822) );
INVX0_RVT U1011 ( .A(n822), .Y(n1023) );
AND2X1_RVT U1012 ( .A1(n1070), .A2(\memory[13][6] ), .Y(n836) );
AND3X1_RVT U1013 ( .A1(n1493), .A2(n1468), .A3(n828), .Y(n823) );
INVX0_RVT U1014 ( .A(n823), .Y(n1024) );
INVX0_RVT U1015 ( .A(n1024), .Y(n1065) );
AND2X1_RVT U1016 ( .A1(n1065), .A2(\memory[1][6] ), .Y(n835) );
AND3X1_RVT U1017 ( .A1(n1458), .A2(n1493), .A3(n1468), .Y(n824) );
INVX0_RVT U1018 ( .A(n824), .Y(n1025) );
INVX0_RVT U1019 ( .A(n1025), .Y(n1066) );
AND2X1_RVT U1020 ( .A1(n1066), .A2(\memory[5][6] ), .Y(n834) );
AND3X1_RVT U1021 ( .A1(rd_ptr_reg[1]), .A2(n1468), .A3(n828), .Y(n825) );
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INVX0_RVT U1022 (.A(n825), .Y(n1026));
INVX0_RVT U1023 (.A(n1026), .Y(n1067));
AND2X1_RVT U1024 (.A1(n1067), .A2(\memory[3][6]), .Y(n832));
NBUFFX2_RVT U1025 (.A(rd_ptr_reg[1]), .Y(n1455));
AND3X1_RVT U1026 (.A1(n1455), .A2(n1458), .A3(n1468), .Y(n826));
INVX0_RVT U1027 (.A(n826), .Y(n1027));
INVX0_RVT U1028 (.A(n1027), .Y(n1068));
AND2X1_RVT U1029 (.A1(n1068), .A2(\memory[7][6]), .Y(n831));
AND3X1_RVT U1030 (.A1(rd_ptr_reg[3]), .A2(n1455), .A3(n828), .Y(n827));
INVX0_RVT U1031 (.A(n827), .Y(n1028));
INVX0_RVT U1032 (.A(n1028), .Y(n1071));
AND2X1_RVT U1033 (.A1(n1071), .A2(\memory[11][6]), .Y(n830));
INVX0_RVT U1034 (.A(n891), .Y(n1029));
INVX0_RVT U1035 (.A(n1029), .Y(n1069));
AND2X1_RVT U1036 (.A1(n1069), .A2(\memory[9][6]), .Y(n829));
OR4X1_RVT U1037 (.A1(n832), .A2(n831), .A3(n830), .A4(n829), .Y(n833));
OR4X1_RVT U1038 (.A1(n836), .A2(n835), .A3(n834), .A4(n833), .Y(n837));
AND2X1_RVT U1039 (.A1(n1368), .A2(n837), .Y(n852));
INVX0_RVT U1040 (.A(n1024), .Y(n1354));
AND2X1_RVT U1041 (.A1(n1354), .A2(\memory[0][6]), .Y(n847));
AND2X1_RVT U1042 (.A1(n824), .A2(\memory[4][6]), .Y(n846));
INVX0_RVT U1043 (.A(n1026), .Y(n1356));
AND2X1_RVT U1044 (.A1(n825), .A2(\memory[2][6]), .Y(n844));
INVX0_RVT U1045 (.A(n1027), .Y(n1361));
AND2X1_RVT U1046 (.A1(n826), .A2(\memory[6][6]), .Y(n843));
AND2X1_RVT U1047 (.A1(n891), .A2(\memory[8][6]), .Y(n841));
AND2X1_RVT U1048 (.A1(n822), .A2(\memory[12][6]), .Y(n840));

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INVX0_RVT U1049 ( .A(n1028), .Y(n1355) );
AND2X1_RVT U1050 ( .A1(n1355), .A2(\memory[10][6] ), .Y(n839) );
AND3X1_RVT U1051 ( .A1(n1455), .A2(rd_ptr_reg[3]), .A3(rd_ptr_reg[2]), .Y(
    n1376) );
INVX0_RVT U1052 ( .A(n849), .Y(n1334) );
AND2X1_RVT U1053 ( .A1(n1334), .A2(\memory[14][6] ), .Y(n838) );
OR3X1_RVT U1054 ( .A1(n844), .A2(n843), .A3(n842), .Y(n845) );
OR3X1_RVT U1055 ( .A1(n847), .A2(n846), .A3(n845), .Y(n848) );
AND2X1_RVT U1056 ( .A1(n1469), .A2(n848), .Y(n851) );
AND2X1_RVT U1057 ( .A1(n1492), .A2(n1376), .Y(n903) );
NBUFFX2_RVT U1058 ( .A(n903), .Y(n1305) );
AND2X1_RVT U1059 ( .A1(\memory[15][6] ), .A2(n1305), .Y(n850) );
OR3X1_RVT U1060 ( .A1(n852), .A2(n851), .A3(n850), .Y(n853) );
AND2X1_RVT U1061 ( .A1(n1456), .A2(n853), .Y(n854) );
OR2X1_RVT U1062 ( .A1(n855), .A2(n854), .Y(n579) );
AND2X1_RVT U1063 ( .A1(data_out_reg[3]), .A2(n1353), .Y(n881) );
AND2X1_RVT U1064 ( .A1(n1070), .A2(\memory[13][3] ), .Y(n863) );
AND2X1_RVT U1065 ( .A1(n1065), .A2(\memory[1][3] ), .Y(n862) );
AND2X1_RVT U1066 ( .A1(n1066), .A2(\memory[5][3] ), .Y(n861) );
AND2X1_RVT U1067 ( .A1(n1067), .A2(\memory[3][3] ), .Y(n859) );
AND2X1_RVT U1068 ( .A1(n1068), .A2(\memory[7][3] ), .Y(n858) );
AND2X1_RVT U1069 ( .A1(n1071), .A2(\memory[11][3] ), .Y(n857) );
AND2X1_RVT U1070 ( .A1(n1069), .A2(\memory[9][3] ), .Y(n856) );
OR4X1_RVT U1071 ( .A1(n859), .A2(n858), .A3(n857), .A4(n856), .Y(n860) );
OR4X1_RVT U1072 ( .A1(n863), .A2(n862), .A3(n861), .A4(n860), .Y(n864) );
AND2X1_RVT U1073 ( .A1(n1368), .A2(n864), .Y(n878) );
AND2X1_RVT U1074 ( .A1(n1354), .A2(\memory[0][3] ), .Y(n874) );
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AND2X1_RVT U1075 ( .A1(n1372), .A2(\memory[4][3] ), .Y(n873) );
AND2X1_RVT U1076 ( .A1(n1356), .A2(\memory[2][3] ), .Y(n871) );
AND2X1_RVT U1077 ( .A1(n1361), .A2(\memory[6][3] ), .Y(n870) );
AND2X1_RVT U1078 ( .A1(n891), .A2(\memory[8][3] ), .Y(n868) );
AND2X1_RVT U1079 ( .A1(n822), .A2(\memory[12][3] ), .Y(n867) );
AND2X1_RVT U1080 ( .A1(n1355), .A2(\memory[10][3] ), .Y(n866) );
AND2X1_RVT U1081 ( .A1(n1334), .A2(\memory[14][3] ), .Y(n865) );
OR3X1_RVT U1082 ( .A1(n871), .A2(n870), .A3(n869), .Y(n872) );
OR3X1_RVT U1083 ( .A1(n874), .A2(n873), .A3(n872), .Y(n875) );
AND2X1_RVT U1084 ( .A1(n1469), .A2(n875), .Y(n877) );
AND2X1_RVT U1085 ( .A1(\memory[15][3] ), .A2(n1305), .Y(n876) );
OR3X1_RVT U1086 ( .A1(n878), .A2(n877), .A3(n876), .Y(n879) );
AND2X1_RVT U1087 ( .A1(n1456), .A2(n879), .Y(n880) );
OR2X1_RVT U1088 ( .A1(n881), .A2(n880), .Y(n576) );
AND2X1_RVT U1089 ( .A1(data_out_reg[0]), .A2(n749), .Y(n909) );
AND2X1_RVT U1090 ( .A1(n1070), .A2(\memory[13][0] ), .Y(n889) );
AND2X1_RVT U1091 ( .A1(n1065), .A2(\memory[1][0] ), .Y(n888) );
AND2X1_RVT U1092 ( .A1(n1066), .A2(\memory[5][0] ), .Y(n887) );
AND2X1_RVT U1093 ( .A1(n1067), .A2(\memory[3][0] ), .Y(n885) );
AND2X1_RVT U1094 ( .A1(n1068), .A2(\memory[7][0] ), .Y(n884) );
AND2X1_RVT U1095 ( .A1(n1071), .A2(\memory[11][0] ), .Y(n883) );
AND2X1_RVT U1096 ( .A1(n1069), .A2(\memory[9][0] ), .Y(n882) );
OR4X1_RVT U1097 ( .A1(n885), .A2(n884), .A3(n883), .A4(n882), .Y(n886) );
OR4X1_RVT U1098 ( .A1(n889), .A2(n888), .A3(n887), .A4(n886), .Y(n890) );
AND2X1_RVT U1099 ( .A1(n1368), .A2(n890), .Y(n906) );
AND2X1_RVT U1100 ( .A1(n1354), .A2(\memory[0][0] ), .Y(n901) );
AND2X1_RVT U1101 ( .A1(n1066), .A2(\memory[4][0] ), .Y(n900) );
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AND2X1_RVT U1102 ( .A1(n1356), .A2(\memory[2][0] ), .Y(n898) );
AND2X1_RVT U1103 ( .A1(n1361), .A2(\memory[6][0] ), .Y(n897) );
AND2X1_RVT U1104 ( .A1(n891), .A2(\memory[8][0] ), .Y(n895) );
AND2X1_RVT U1105 ( .A1(n822), .A2(\memory[12][0] ), .Y(n894) );
AND2X1_RVT U1106 ( .A1(n1355), .A2(\memory[10][0] ), .Y(n893) );
AND2X1_RVT U1107 ( .A1(n1334), .A2(\memory[14][0] ), .Y(n892) );
OR4X1_RVT U1108 ( .A1(n895), .A2(n894), .A3(n893), .A4(n892), .Y(n896) );
OR3X1_RVT U1109 ( .A1(n898), .A2(n897), .A3(n896), .Y(n899) );
OR3X1_RVT U1110 ( .A1(n901), .A2(n900), .A3(n899), .Y(n902) );
AND2X1_RVT U1111 ( .A1(n1469), .A2(n902), .Y(n905) );
NBUFFX2_RVT U1112 ( .A(n903), .Y(n1370) );
AND2X1_RVT U1113 ( .A1(\memory[15][0] ), .A2(n1370), .Y(n904) );
OR3X1_RVT U1114 ( .A1(n906), .A2(n905), .A3(n904), .Y(n907) );
AND2X1_RVT U1115 ( .A1(n1456), .A2(n907), .Y(n908) );
OR2X1_RVT U1116 ( .A1(n909), .A2(n908), .Y(n573) );
INVX0_RVT U1117 ( .A(n911), .Y(n1491) );
INVX0_RVT U1118 ( .A(n911), .Y(n1490) );
INVX0_RVT U1119 ( .A(n911), .Y(n1487) );
OR2X1_RVT U1120 ( .A1(n912), .A2(n1471), .Y(n1446) );
AND2X1_RVT U1121 ( .A1(n1446), .A2(n1479), .Y(n913) );
OR2X1_RVT U1122 ( .A1(n1447), .A2(n913), .Y(n914) );
INVX0_RVT U1123 ( .A(n914), .Y(n587) );
AND3X1_RVT U1124 ( .A1(n1321), .A2(n1368), .A3(n1455), .Y(n1457) );
OR2X1_RVT U1125 ( .A1(n1353), .A2(n1346), .Y(n1454) );
AND2X1_RVT U1126 ( .A1(n1493), .A2(n1454), .Y(n915) );
OR2X1_RVT U1127 ( .A1(n1457), .A2(n915), .Y(n916) );
AND2X1_RVT U1128 ( .A1(count_reg[0]), .A2(n917), .Y(n921) );
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AND2X1_RVT U1129 (.A1(n919), .A2(n918), .Y(n920));

OR2X1_RVT U1130 (.A1(n921), .A2(n920), .Y(n922));

OR2X1_RVT U1131 (.A1(n1481), .A2(n1426), .Y(n924));

AND2X1_RVT U1132 (.A1(n1281), .A2(n1426), .Y(n925));

OR2X1_RVT U1133 (.A1(n926), .A2(n925), .Y(n656));

OR2X1_RVT U1134 (.A1(n1449), .A2(n1482), .Y(n1452));

AND2X1_RVT U1135 (.A1(\memory[15][7]), .A2(n1013), .Y(n930));

OR2X1_RVT U1136 (.A1(n1008), .A2(n1486), .Y(n928));

OR2X1_RVT U1137 (.A1(n930), .A2(n929), .Y(n592));

AND2X1_RVT U1138 (.A1(\memory[15][6]), .A2(n1008), .Y(n934));

OR2X1_RVT U1139 (.A1(n1013), .A2(n1485), .Y(n932));

OR2X1_RVT U1140 (.A1(n934), .A2(n933), .Y(n593));

AND2X1_RVT U1141 (.A1(\memory[2][4]), .A2(n940), .Y(n937));

AND2X1_RVT U1142 (.A1(data_in_reg[4]), .A2(n1003), .Y(n936));

OR2X1_RVT U1143 (.A1(n937), .A2(n936), .Y(n699));

AND2X1_RVT U1144 (.A1(\memory[2][1]), .A2(n940), .Y(n939));

AND2X1_RVT U1145 (.A1(data_in_reg[1]), .A2(n1000), .Y(n938));

OR2X1_RVT U1146 (.A1(n939), .A2(n938), .Y(n702));

AND2X1_RVT U1147 (.A1(\memory[2][7]), .A2(n940), .Y(n943));

OR2X1_RVT U1148 (.A1(n1486), .A2(n940), .Y(n941));

OR2X1_RVT U1149 (.A1(n943), .A2(n942), .Y(n696));

AND2X1_RVT U1150 (.A1(count_reg[2]), .A2(n944), .Y(n948));

AND2X1_RVT U1151 (.A1(n946), .A2(n945), .Y(n947));

OR2X1_RVT U1152 (.A1(n948), .A2(n947), .Y(n590));

INVX0_RVT U1153 (.A(n949), .Y(n1100));

OR3X1_RVT U1154 (.A1(n1482), .A2(n972), .A3(n1100), .Y(n950));

AND2X1_RVT U1155 (.A1(\memory[13][1]), .A2(n965), .Y(n952));

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NBUFFX2_RVT U1156 ( .A(n962), .Y(n969) );
AND2X1_RVT U1157 ( .A1(data_in_reg[1]), .A2(n969), .Y(n951) );
OR2X1_RVT U1158 ( .A1(n952), .A2(n951), .Y(n614) );
AND2X1_RVT U1159 ( .A1(\memory[13][7] ), .A2(n965), .Y(n954) );
AND2X1_RVT U1160 ( .A1(n1281), .A2(n969), .Y(n953) );
OR2X1_RVT U1161 ( .A1(n954), .A2(n953), .Y(n608) );
AND2X1_RVT U1162 ( .A1(\memory[13][3] ), .A2(n968), .Y(n957) );
AND2X1_RVT U1163 ( .A1(n1415), .A2(n955), .Y(n956) );
OR2X1_RVT U1164 ( .A1(n957), .A2(n956), .Y(n612) );
AND2X1_RVT U1165 ( .A1(\memory[13][5] ), .A2(n965), .Y(n959) );
AND2X1_RVT U1166 ( .A1(n1431), .A2(n962), .Y(n958) );
OR2X1_RVT U1167 ( .A1(n959), .A2(n958), .Y(n610) );
AND2X1_RVT U1168 ( .A1(\memory[13][2] ), .A2(n968), .Y(n961) );
AND2X1_RVT U1169 ( .A1(data_in_reg[2]), .A2(n969), .Y(n960) );
OR2X1_RVT U1170 ( .A1(n961), .A2(n960), .Y(n613) );
AND2X1_RVT U1171 ( .A1(\memory[13][6] ), .A2(n968), .Y(n964) );
AND2X1_RVT U1172 ( .A1(n1277), .A2(n962), .Y(n963) );
OR2X1_RVT U1173 ( .A1(n964), .A2(n963), .Y(n609) );
AND2X1_RVT U1174 ( .A1(\memory[13][4] ), .A2(n965), .Y(n967) );
AND2X1_RVT U1175 ( .A1(n1195), .A2(n962), .Y(n966) );
OR2X1_RVT U1176 ( .A1(n967), .A2(n966), .Y(n611) );
AND2X1_RVT U1177 ( .A1(\memory[13][0] ), .A2(n968), .Y(n971) );
AND2X1_RVT U1178 ( .A1(n1293), .A2(n969), .Y(n970) );
OR2X1_RVT U1179 ( .A1(n971), .A2(n970), .Y(n615) );
OR2X1_RVT U1180 ( .A1(n791), .A2(n972), .Y(n977) );
NBUFFX2_RVT U1181 ( .A(n974), .Y(n987) );
AND2X1_RVT U1182 ( .A1(\memory[1][5] ), .A2(n987), .Y(n976) );
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AND2X1_RVT U1183 ( .A1(n1431), .A2(n994), .Y(n975) );
OR2X1_RVT U1184 ( .A1(n976), .A2(n975), .Y(n706) );
AND2X1_RVT U1185 ( .A1(\memory[1][4] ), .A2(n993), .Y(n979) );
AND2X1_RVT U1186 ( .A1(data_in_reg[4]), .A2(n988), .Y(n978) );
OR2X1_RVT U1187 ( .A1(n979), .A2(n978), .Y(n707) );
AND2X1_RVT U1188 ( .A1(\memory[1][3] ), .A2(n993), .Y(n982) );
AND2X1_RVT U1189 ( .A1(data_in_reg[3]), .A2(n988), .Y(n981) );
OR2X1_RVT U1190 ( .A1(n982), .A2(n981), .Y(n708) );
AND2X1_RVT U1191 ( .A1(\memory[1][6] ), .A2(n987), .Y(n984) );
AND2X1_RVT U1192 ( .A1(n1277), .A2(n988), .Y(n983) );
OR2X1_RVT U1193 ( .A1(n984), .A2(n983), .Y(n705) );
AND2X1_RVT U1194 ( .A1(\memory[1][7] ), .A2(n993), .Y(n986) );
AND2X1_RVT U1195 ( .A1(n1281), .A2(n988), .Y(n985) );
OR2X1_RVT U1196 ( .A1(n986), .A2(n985), .Y(n704) );
AND2X1_RVT U1197 ( .A1(\memory[1][2] ), .A2(n987), .Y(n990) );
AND2X1_RVT U1198 ( .A1(n1244), .A2(n988), .Y(n989) );
OR2X1_RVT U1199 ( .A1(n990), .A2(n989), .Y(n709) );
AND2X1_RVT U1200 ( .A1(\memory[1][0] ), .A2(n974), .Y(n992) );
AND2X1_RVT U1201 ( .A1(n1293), .A2(n994), .Y(n991) );
OR2X1_RVT U1202 ( .A1(n992), .A2(n991), .Y(n711) );
AND2X1_RVT U1203 ( .A1(\memory[1][1] ), .A2(n993), .Y(n996) );
AND2X1_RVT U1204 ( .A1(n1239), .A2(n994), .Y(n995) );
OR2X1_RVT U1205 ( .A1(n996), .A2(n995), .Y(n710) );
AND2X1_RVT U1206 ( .A1(\memory[2][5] ), .A2(n792), .Y(n999) );
AND2X1_RVT U1207 ( .A1(data_in_reg[5]), .A2(n1003), .Y(n998) );
OR2X1_RVT U1208 ( .A1(n999), .A2(n998), .Y(n698) );
AND2X1_RVT U1209 ( .A1(\memory[2][2] ), .A2(n997), .Y(n1002) );
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AND2X1_RVT U1210 ( .A1(data_in_reg[2]), .A2(n1000), .Y(n1001) );
OR2X1_RVT U1211 ( .A1(n1002), .A2(n1001), .Y(n701) );
AND2X1_RVT U1212 ( .A1(\memory[2][3] ), .A2(n997), .Y(n1005) );
AND2X1_RVT U1213 ( .A1(data_in_reg[3]), .A2(n1003), .Y(n1004) );
OR2X1_RVT U1214 ( .A1(n1005), .A2(n1004), .Y(n700) );
AND2X1_RVT U1215 ( .A1(\memory[15][1] ), .A2(n1008), .Y(n1007) );
AND2X1_RVT U1216 ( .A1(n1020), .A2(n1239), .Y(n1006) );
OR2X1_RVT U1217 ( .A1(n1007), .A2(n1006), .Y(n598) );
AND2X1_RVT U1218 ( .A1(\memory[15][4] ), .A2(n1008), .Y(n1012) );
OR2X1_RVT U1219 ( .A1(n1009), .A2(n1474), .Y(n1010) );
INVX0_RVT U1220 ( .A(n1010), .Y(n1011) );
OR2X1_RVT U1221 ( .A1(n1012), .A2(n1011), .Y(n595) );
AND2X1_RVT U1222 ( .A1(\memory[15][0] ), .A2(n1013), .Y(n1015) );
AND2X1_RVT U1223 ( .A1(n1020), .A2(n1235), .Y(n1014) );
OR2X1_RVT U1224 ( .A1(n1015), .A2(n1014), .Y(n599) );
AND2X1_RVT U1225 ( .A1(\memory[15][2] ), .A2(n1013), .Y(n1017) );
AND2X1_RVT U1226 ( .A1(n1020), .A2(n1244), .Y(n1016) );
OR2X1_RVT U1227 ( .A1(n1017), .A2(n1016), .Y(n597) );
AND2X1_RVT U1228 ( .A1(\memory[15][3] ), .A2(n1013), .Y(n1019) );
AND2X1_RVT U1229 ( .A1(n1020), .A2(n1415), .Y(n1018) );
OR2X1_RVT U1230 ( .A1(n1019), .A2(n1018), .Y(n596) );
AND2X1_RVT U1231 ( .A1(\memory[15][5] ), .A2(n1008), .Y(n1022) );
AND2X1_RVT U1232 ( .A1(n1020), .A2(n1187), .Y(n1021) );
OR2X1_RVT U1233 ( .A1(n1022), .A2(n1021), .Y(n594) );
AND2X1_RVT U1234 ( .A1(data_out_reg[5]), .A2(n1324), .Y(n1055) );
INVX0_RVT U1235 ( .A(n1023), .Y(n1374) );
AND2X1_RVT U1236 ( .A1(n1374), .A2(\memory[13][5] ), .Y(n1037) );
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INVX0_RVT U1237 ( .A(n1024), .Y(n1371) );
AND2X1_RVT U1238 ( .A1(n1371), .A2(\memory[1][5] ), .Y(n1036) );
INVX0_RVT U1239 ( .A(n1025), .Y(n1372) );
AND2X1_RVT U1240 ( .A1(n1372), .A2(\memory[5][5] ), .Y(n1035) );
AND2X1_RVT U1241 ( .A1(n825), .A2(\memory[3][5] ), .Y(n1033) );
AND2X1_RVT U1242 ( .A1(n826), .A2(\memory[7][5] ), .Y(n1032) );
INVX0_RVT U1243 ( .A(n1028), .Y(n1375) );
AND2X1_RVT U1244 ( .A1(n1375), .A2(\memory[11][5] ), .Y(n1031) );
INVX0_RVT U1245 ( .A(n1029), .Y(n1373) );
AND2X1_RVT U1246 ( .A1(n1373), .A2(\memory[9][5] ), .Y(n1030) );
OR4X1_RVT U1247 ( .A1(n1033), .A2(n1032), .A3(n1031), .A4(n1030), .Y(n1034)
    );
OR4X1_RVT U1248 ( .A1(n1037), .A2(n1036), .A3(n1035), .A4(n1034), .Y(n1038)
    );
AND2X1_RVT U1249 ( .A1(n1492), .A2(n1038), .Y(n1052) );
AND2X1_RVT U1250 ( .A1(n1065), .A2(\memory[0][5] ), .Y(n1048) );
AND2X1_RVT U1251 ( .A1(n1066), .A2(\memory[4][5] ), .Y(n1047) );
AND2X1_RVT U1252 ( .A1(n1067), .A2(\memory[2][5] ), .Y(n1045) );
AND2X1_RVT U1253 ( .A1(n1068), .A2(\memory[6][5] ), .Y(n1044) );
AND2X1_RVT U1254 ( .A1(n1069), .A2(\memory[8][5] ), .Y(n1042) );
AND2X1_RVT U1255 ( .A1(n1070), .A2(\memory[12][5] ), .Y(n1041) );
AND2X1_RVT U1256 ( .A1(n1071), .A2(\memory[10][5] ), .Y(n1040) );
AND2X1_RVT U1257 ( .A1(n1334), .A2(\memory[14][5] ), .Y(n1039) );
OR4X1_RVT U1258 ( .A1(n1042), .A2(n1041), .A3(n1040), .A4(n1039), .Y(n1043)
    );
OR3X1_RVT U1259 ( .A1(n1045), .A2(n1044), .A3(n1043), .Y(n1046) );
OR3X1_RVT U1260 ( .A1(n1048), .A2(n1047), .A3(n1046), .Y(n1049) );
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AND2X1_RVT U1261 ( .A1(n1346), .A2(n1049), .Y(n1051) );
AND2X1_RVT U1262 ( .A1(\memory[15][5] ), .A2(n1370), .Y(n1050) );
OR3X1_RVT U1263 ( .A1(n1052), .A2(n1051), .A3(n1050), .Y(n1053) );
AND2X1_RVT U1264 ( .A1(n1321), .A2(n1053), .Y(n1054) );
OR2X1_RVT U1265 ( .A1(n1055), .A2(n1054), .Y(n578) );
AND2X1_RVT U1266 ( .A1(data_out_reg[2]), .A2(n1324), .Y(n1088) );
AND2X1_RVT U1267 ( .A1(n1374), .A2(\memory[13][2] ), .Y(n1063) );
AND2X1_RVT U1268 ( .A1(n1371), .A2(\memory[1][2] ), .Y(n1062) );
AND2X1_RVT U1269 ( .A1(n1372), .A2(\memory[5][2] ), .Y(n1061) );
AND2X1_RVT U1270 ( .A1(n825), .A2(\memory[3][2] ), .Y(n1059) );
AND2X1_RVT U1271 ( .A1(n826), .A2(\memory[7][2] ), .Y(n1058) );
AND2X1_RVT U1272 ( .A1(n1375), .A2(\memory[11][2] ), .Y(n1057) );
AND2X1_RVT U1273 ( .A1(n1373), .A2(\memory[9][2] ), .Y(n1056) );
OR4X1_RVT U1274 ( .A1(n1059), .A2(n1058), .A3(n1057), .A4(n1056), .Y(n1060)
    );
OR4X1_RVT U1275 ( .A1(n1063), .A2(n1062), .A3(n1061), .A4(n1060), .Y(n1064)
    );
AND2X1_RVT U1276 ( .A1(n1492), .A2(n1064), .Y(n1085) );
AND2X1_RVT U1277 ( .A1(n1065), .A2(\memory[0][2] ), .Y(n1081) );
AND2X1_RVT U1278 ( .A1(n1066), .A2(\memory[4][2] ), .Y(n1080) );
AND2X1_RVT U1279 ( .A1(n1067), .A2(\memory[2][2] ), .Y(n1078) );
AND2X1_RVT U1280 ( .A1(n1068), .A2(\memory[6][2] ), .Y(n1077) );
AND2X1_RVT U1281 ( .A1(n1069), .A2(\memory[8][2] ), .Y(n1075) );
AND2X1_RVT U1282 ( .A1(n1070), .A2(\memory[12][2] ), .Y(n1074) );
AND2X1_RVT U1283 ( .A1(n1071), .A2(\memory[10][2] ), .Y(n1073) );
AND2X1_RVT U1284 ( .A1(n1376), .A2(\memory[14][2] ), .Y(n1072) );
OR4X1_RVT U1285 ( .A1(n1075), .A2(n1074), .A3(n1073), .A4(n1072), .Y(n1076)
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    );
OR3X1_RVT U1286 ( .A1(n1078), .A2(n1077), .A3(n1076), .Y(n1079) );
OR3X1_RVT U1287 ( .A1(n1081), .A2(n1080), .A3(n1079), .Y(n1082) );
AND2X1_RVT U1288 ( .A1(n1346), .A2(n1082), .Y(n1084) );
AND2X1_RVT U1289 ( .A1(\memory[15][2] ), .A2(n1305), .Y(n1083) );
OR3X1_RVT U1290 ( .A1(n1085), .A2(n1084), .A3(n1083), .Y(n1086) );
AND2X1_RVT U1291 ( .A1(n1321), .A2(n1086), .Y(n1087) );
OR2X1_RVT U1292 ( .A1(n1088), .A2(n1087), .Y(n575) );
AND2X1_RVT U1293 ( .A1(\memory[0][7] ), .A2(n1093), .Y(n1090) );
AND2X1_RVT U1294 ( .A1(data_in_reg[7]), .A2(n1097), .Y(n1089) );
OR2X1_RVT U1295 ( .A1(n1090), .A2(n1089), .Y(n713) );
AND2X1_RVT U1296 ( .A1(\memory[0][5] ), .A2(n1093), .Y(n1092) );
AND2X1_RVT U1297 ( .A1(n1187), .A2(n1097), .Y(n1091) );
OR2X1_RVT U1298 ( .A1(n1092), .A2(n1091), .Y(n715) );
AND2X1_RVT U1299 ( .A1(\memory[0][4] ), .A2(n1093), .Y(n1095) );
AND2X1_RVT U1300 ( .A1(n1195), .A2(n1097), .Y(n1094) );
OR2X1_RVT U1301 ( .A1(n1095), .A2(n1094), .Y(n716) );
NBUFFX2_RVT U1302 ( .A(n1096), .Y(n1148) );
AND2X1_RVT U1303 ( .A1(\memory[0][0] ), .A2(n1148), .Y(n1099) );
AND2X1_RVT U1304 ( .A1(n1235), .A2(n1097), .Y(n1098) );
OR2X1_RVT U1305 ( .A1(n1099), .A2(n1098), .Y(n712) );
AND2X1_RVT U1306 ( .A1(\memory[14][3] ), .A2(n1161), .Y(n1104) );
AND2X1_RVT U1307 ( .A1(data_in_reg[3]), .A2(n1123), .Y(n1103) );
OR2X1_RVT U1308 ( .A1(n1104), .A2(n1103), .Y(n604) );
AND2X1_RVT U1309 ( .A1(\memory[14][6] ), .A2(n1161), .Y(n1106) );
AND2X1_RVT U1310 ( .A1(data_in_reg[6]), .A2(n1123), .Y(n1105) );
OR2X1_RVT U1311 ( .A1(n1106), .A2(n1105), .Y(n601) );
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AND2X1_RVT U1312 (.A1(\memory[14][7]), .A2(n1166), .Y(n1108));
AND2X1_RVT U1313 (.A1(data_in_reg[7]), .A2(n1123), .Y(n1107));
OR2X1_RVT U1314 (.A1(n1108), .A2(n1107), .Y(n600));
AND2X1_RVT U1315 (.A1(\memory[14][4]), .A2(n1166), .Y(n1110));
AND2X1_RVT U1316 (.A1(n1195), .A2(n1123), .Y(n1109));
OR2X1_RVT U1317 (.A1(n1110), .A2(n1109), .Y(n603));
OR3X2_RVT U1318 (.A1(n1111), .A2(n1268), .A3(n1482), .Y(n1112));
AND2X1_RVT U1319 (.A1(\memory[12][5]), .A2(n1157), .Y(n1114));
AND2X1_RVT U1320 (.A1(n1187), .A2(n1126), .Y(n1113));
OR2X1_RVT U1321 (.A1(n1114), .A2(n1113), .Y(n618));
AND2X1_RVT U1322 (.A1(\memory[12][7]), .A2(n1157), .Y(n1116));
AND2X1_RVT U1323 (.A1(data_in_reg[7]), .A2(n1126), .Y(n1115));
OR2X1_RVT U1324 (.A1(n1116), .A2(n1115), .Y(n616));
AND2X1_RVT U1325 (.A1(\memory[12][3]), .A2(n1152), .Y(n1119));
AND2X1_RVT U1326 (.A1(n1415), .A2(n1126), .Y(n1118));
OR2X1_RVT U1327 (.A1(n1119), .A2(n1118), .Y(n620));
AND2X1_RVT U1328 (.A1(\memory[12][6]), .A2(n1152), .Y(n1121));
AND2X1_RVT U1329 (.A1(data_in_reg[6]), .A2(n1126), .Y(n1120));
OR2X1_RVT U1330 (.A1(n1121), .A2(n1120), .Y(n617));
AND2X1_RVT U1331 (.A1(\memory[14][5]), .A2(n1166), .Y(n1125));
AND2X1_RVT U1332 (.A1(data_in_reg[5]), .A2(n1123), .Y(n1124));
OR2X1_RVT U1333 (.A1(n1125), .A2(n1124), .Y(n602));
AND2X1_RVT U1334 (.A1(\memory[12][4]), .A2(n1152), .Y(n1128));
AND2X1_RVT U1335 (.A1(n1195), .A2(n1126), .Y(n1127));
OR2X1_RVT U1336 (.A1(n1128), .A2(n1127), .Y(n619));
AND2X1_RVT U1337 (.A1(\memory[3][4]), .A2(n1176), .Y(n1132));
AND2X1_RVT U1338 (.A1(n1195), .A2(n1140), .Y(n1131));

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OR2X1_RVT U1339 ( .A1(n1132), .A2(n1131), .Y(n688) );
AND2X1_RVT U1340 ( .A1(\memory[3][3] ), .A2(n1173), .Y(n1134) );
AND2X1_RVT U1341 ( .A1(n1415), .A2(n1140), .Y(n1133) );
OR2X1_RVT U1342 ( .A1(n1134), .A2(n1133), .Y(n689) );
AND2X1_RVT U1343 ( .A1(\memory[3][2] ), .A2(n1170), .Y(n1137) );
AND2X1_RVT U1344 ( .A1(n1244), .A2(n1140), .Y(n1136) );
OR2X1_RVT U1345 ( .A1(n1137), .A2(n1136), .Y(n690) );
AND2X1_RVT U1346 ( .A1(\memory[3][1] ), .A2(n1176), .Y(n1139) );
AND2X1_RVT U1347 ( .A1(n1239), .A2(n1140), .Y(n1138) );
OR2X1_RVT U1348 ( .A1(n1139), .A2(n1138), .Y(n691) );
AND2X1_RVT U1349 ( .A1(\memory[3][0] ), .A2(n1173), .Y(n1142) );
AND2X1_RVT U1350 ( .A1(n1235), .A2(n1140), .Y(n1141) );
OR2X1_RVT U1351 ( .A1(n1142), .A2(n1141), .Y(n692) );
AND2X1_RVT U1352 ( .A1(\memory[0][2] ), .A2(n1148), .Y(n1145) );
INVX0_RVT U1353 ( .A(n1096), .Y(n1149) );
AND2X1_RVT U1354 ( .A1(n1244), .A2(n1149), .Y(n1144) );
OR2X1_RVT U1355 ( .A1(n1145), .A2(n1144), .Y(n718) );
AND2X1_RVT U1356 ( .A1(\memory[0][3] ), .A2(n1093), .Y(n1147) );
AND2X1_RVT U1357 ( .A1(n1415), .A2(n1149), .Y(n1146) );
OR2X1_RVT U1358 ( .A1(n1147), .A2(n1146), .Y(n717) );
AND2X1_RVT U1359 ( .A1(\memory[0][1] ), .A2(n1148), .Y(n1151) );
AND2X1_RVT U1360 ( .A1(n1239), .A2(n1149), .Y(n1150) );
OR2X1_RVT U1361 ( .A1(n1151), .A2(n1150), .Y(n719) );
AND2X1_RVT U1362 ( .A1(\memory[12][0] ), .A2(n1157), .Y(n1154) );
AND2X1_RVT U1363 ( .A1(n1235), .A2(n1158), .Y(n1153) );
OR2X1_RVT U1364 ( .A1(n1154), .A2(n1153), .Y(n623) );
AND2X1_RVT U1365 ( .A1(\memory[12][2] ), .A2(n1157), .Y(n1156) );
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AND2X1_RVT U1366 (.A1(n1244), .A2(n1158), .Y(n1155));
OR2X1_RVT U1367 (.A1(n1156), .A2(n1155), .Y(n621));
AND2X1_RVT U1368 (.A1(\memory[12][1]), .A2(n1157), .Y(n1160));
AND2X1_RVT U1369 (.A1(n1239), .A2(n1158), .Y(n1159));
OR2X1_RVT U1370 (.A1(n1160), .A2(n1159), .Y(n622));
AND2X1_RVT U1371 (.A1(\memory[14][0]), .A2(n1161), .Y(n1163));
AND2X1_RVT U1372 (.A1(n1293), .A2(n1167), .Y(n1162));
OR2X1_RVT U1373 (.A1(n1163), .A2(n1162), .Y(n607));
AND2X1_RVT U1374 (.A1(\memory[14][1]), .A2(n1166), .Y(n1165));
AND2X1_RVT U1375 (.A1(data_in_reg[1]), .A2(n1167), .Y(n1164));
OR2X1_RVT U1376 (.A1(n1165), .A2(n1164), .Y(n606));
AND2X1_RVT U1377 (.A1(\memory[14][2]), .A2(n1166), .Y(n1169));
AND2X1_RVT U1378 (.A1(data_in_reg[2]), .A2(n1167), .Y(n1168));
OR2X1_RVT U1379 (.A1(n1169), .A2(n1168), .Y(n605));
AND2X1_RVT U1380 (.A1(\memory[3][7]), .A2(n1170), .Y(n1172));
AND2X1_RVT U1381 (.A1(data_in_reg[7]), .A2(n1177), .Y(n1171));
OR2X1_RVT U1382 (.A1(n1172), .A2(n1171), .Y(n693));
AND2X1_RVT U1383 (.A1(\memory[3][5]), .A2(n1173), .Y(n1175));
AND2X1_RVT U1384 (.A1(n1187), .A2(n1177), .Y(n1174));
OR2X1_RVT U1385 (.A1(n1175), .A2(n1174), .Y(n695));
AND2X1_RVT U1386 (.A1(\memory[3][6]), .A2(n1176), .Y(n1179));
AND2X1_RVT U1387 (.A1(data_in_reg[6]), .A2(n1177), .Y(n1178));
OR2X1_RVT U1388 (.A1(n1179), .A2(n1178), .Y(n694));
INVX0_RVT U1389 (.A(n1200), .Y(n1247));
AND2X1_RVT U1390 (.A1(\memory[10][5]), .A2(n1247), .Y(n1181));
AND2X1_RVT U1391 (.A1(n1431), .A2(n728), .Y(n1180));
OR2X1_RVT U1392 (.A1(n1181), .A2(n1180), .Y(n634));

AND2X1_RVT U1393 (.A1(\memory[8][4]), .A2(n1230), .Y(n1184));
AND2X1_RVT U1394 (.A1(data_in_reg[4]), .A2(n1208), .Y(n1183));
AND2X1_RVT U1395 (.A1(\memory[8][3]), .A2(n1224), .Y(n1186));
AND2X1_RVT U1396 (.A1(data_in_reg[3]), .A2(n1208), .Y(n1185));
OR2X1_RVT U1397 (.A1(n1186), .A2(n1185), .Y(n652));
INVX0_RVT U1398 (.A(n1192), .Y(n1234));
AND2X1_RVT U1399 (.A1(\memory[9][5]), .A2(n1234), .Y(n1189));
AND2X1_RVT U1400 (.A1(n1187), .A2(n1205), .Y(n1188));
OR2X1_RVT U1401 (.A1(n1189), .A2(n1188), .Y(n642));
AND2X1_RVT U1402 (.A1(\memory[8][5]), .A2(n1230), .Y(n1191));
AND2X1_RVT U1403 (.A1(data_in_reg[5]), .A2(n1208), .Y(n1190));
AND2X1_RVT U1404 (.A1(\memory[9][3]), .A2(n1238), .Y(n1194));
AND2X1_RVT U1405 (.A1(n1415), .A2(n1205), .Y(n1193));
OR2X1_RVT U1406 (.A1(n1194), .A2(n1193), .Y(n644));
AND2X1_RVT U1407 (.A1(\memory[9][4]), .A2(n1242), .Y(n1197));
AND2X1_RVT U1408 (.A1(n1195), .A2(n1205), .Y(n1196));
AND2X1_RVT U1409 (.A1(\memory[10][4]), .A2(n763), .Y(n1199));
AND2X1_RVT U1410 (.A1(data_in_reg[4]), .A2(n728), .Y(n1198));
OR2X1_RVT U1411 (.A1(n1199), .A2(n1198), .Y(n635));
AND2X1_RVT U1412 (.A1(\memory[10][3]), .A2(n1252), .Y(n1201));
AND2X1_RVT U1413 (.A1(\memory[10][6]), .A2(n1252), .Y(n1204));
AND2X1_RVT U1414 (.A1(n1277), .A2(n728), .Y(n1203));
OR2X1_RVT U1415 (.A1(n1204), .A2(n1203), .Y(n633));
AND2X1_RVT U1416 (.A1(\memory[9][6]), .A2(n1238), .Y(n1207));
AND2X1_RVT U1417 (.A1(data_in_reg[6]), .A2(n1205), .Y(n1206));
OR2X1_RVT U1418 (.A1(n1207), .A2(n1206), .Y(n641));
AND2X1_RVT U1419 (.A1(\memory[8][6]), .A2(n1224), .Y(n1210));

AND2X1_RVT U1420 (.A1(data_in_reg[6]), .A2(n1208), .Y(n1209));
OR2X1_RVT U1421 (.A1(n1210), .A2(n1209), .Y(n649));
AND2X1_RVT U1422 (.A1(\memory[11][3]), .A2(n1262), .Y(n1214));
AND2X1_RVT U1423 (.A1(data_in_reg[3]), .A2(n1221), .Y(n1213));
OR2X1_RVT U1424 (.A1(n1214), .A2(n1213), .Y(n628));
AND2X1_RVT U1425 (.A1(\memory[11][5]), .A2(n1212), .Y(n1216));
AND2X1_RVT U1426 (.A1(data_in_reg[5]), .A2(n1221), .Y(n1215));
OR2X1_RVT U1427 (.A1(n1216), .A2(n1215), .Y(n626));
AND2X1_RVT U1428 (.A1(\memory[11][6]), .A2(n1262), .Y(n1218));
AND2X1_RVT U1429 (.A1(n1277), .A2(n1221), .Y(n1217));
OR2X1_RVT U1430 (.A1(n1218), .A2(n1217), .Y(n625));
AND2X1_RVT U1431 (.A1(\memory[11][4]), .A2(n1262), .Y(n1220));
AND2X1_RVT U1432 (.A1(data_in_reg[4]), .A2(n1221), .Y(n1219));
OR2X1_RVT U1433 (.A1(n1220), .A2(n1219), .Y(n627));
AND2X1_RVT U1434 (.A1(\memory[11][7]), .A2(n1259), .Y(n1223));
AND2X1_RVT U1435 (.A1(data_in_reg[7]), .A2(n1221), .Y(n1222));
OR2X1_RVT U1436 (.A1(n1223), .A2(n1222), .Y(n624));
AND2X1_RVT U1437 (.A1(\memory[8][1]), .A2(n1224), .Y(n1227));
AND2X1_RVT U1438 (.A1(data_in_reg[1]), .A2(n1231), .Y(n1226));
OR2X1_RVT U1439 (.A1(n1227), .A2(n1226), .Y(n654));
AND2X1_RVT U1440 (.A1(\memory[8][2]), .A2(n1224), .Y(n1229));
AND2X1_RVT U1441 (.A1(data_in_reg[2]), .A2(n1231), .Y(n1228));
OR2X1_RVT U1442 (.A1(n1229), .A2(n1228), .Y(n653));
AND2X1_RVT U1443 (.A1(\memory[8][0]), .A2(n1230), .Y(n1233));
AND2X1_RVT U1444 (.A1(data_in_reg[0]), .A2(n1231), .Y(n1232));
AND2X1_RVT U1445 (.A1(\memory[9][0]), .A2(n1242), .Y(n1237));
AND2X1_RVT U1446 (.A1(n1235), .A2(n1243), .Y(n1236));

AND2X1_RVT U1447 (.A1(\memory[9][1]), .A2(n1238), .Y(n1241));
AND2X1_RVT U1448 (.A1(n1239), .A2(n1243), .Y(n1240));
OR2X1_RVT U1449 (.A1(n1241), .A2(n1240), .Y(n646));
AND2X1_RVT U1450 (.A1(\memory[9][2]), .A2(n1242), .Y(n1246));
AND2X1_RVT U1451 (.A1(n1244), .A2(n1243), .Y(n1245));
AND2X1_RVT U1452 (.A1(\memory[10][0]), .A2(n1252), .Y(n1249));
AND2X1_RVT U1453 (.A1(n1293), .A2(n1253), .Y(n1248));
OR2X1_RVT U1454 (.A1(n1249), .A2(n1248), .Y(n639));
AND2X1_RVT U1455 (.A1(\memory[10][2]), .A2(n763), .Y(n1251));
AND2X1_RVT U1456 (.A1(data_in_reg[2]), .A2(n1253), .Y(n1250));
OR2X1_RVT U1457 (.A1(n1251), .A2(n1250), .Y(n637));
AND2X1_RVT U1458 (.A1(\memory[10][1]), .A2(n1252), .Y(n1255));
AND2X1_RVT U1459 (.A1(data_in_reg[1]), .A2(n1253), .Y(n1254));
OR2X1_RVT U1460 (.A1(n1255), .A2(n1254), .Y(n638));
AND2X1_RVT U1461 (.A1(\memory[11][1]), .A2(n1259), .Y(n1258));
AND2X1_RVT U1462 (.A1(data_in_reg[1]), .A2(n1263), .Y(n1257));
OR2X1_RVT U1463 (.A1(n1258), .A2(n1257), .Y(n630));
AND2X1_RVT U1464 (.A1(\memory[11][2]), .A2(n1259), .Y(n1261));
AND2X1_RVT U1465 (.A1(data_in_reg[2]), .A2(n1263), .Y(n1260));
OR2X1_RVT U1466 (.A1(n1261), .A2(n1260), .Y(n629));
AND2X1_RVT U1467 (.A1(\memory[11][0]), .A2(n1262), .Y(n1265));
AND2X1_RVT U1468 (.A1(data_in_reg[0]), .A2(n1263), .Y(n1264));
OR2X1_RVT U1469 (.A1(n1265), .A2(n1264), .Y(n631));
NBUFFX2_RVT U1470 (.A(n1266), .Y(n1267));
AND2X1_RVT U1471 (.A1(\memory[4][5]), .A2(n1291), .Y(n1271));
AND2X1_RVT U1472 (.A1(n1431), .A2(n1280), .Y(n1270));
OR2X1_RVT U1473 (.A1(n1271), .A2(n1270), .Y(n682));

AND2X1_RVT U1474 (.A1(\memory[4][3]), .A2(n733), .Y(n1273));
AND2X1_RVT U1475 (.A1(data_in_reg[3]), .A2(n1280), .Y(n1272));
AND2X1_RVT U1476 (.A1(\memory[4][4]), .A2(n1291), .Y(n1276));
AND2X1_RVT U1477 (.A1(data_in_reg[4]), .A2(n1280), .Y(n1275));
OR2X1_RVT U1478 (.A1(n1276), .A2(n1275), .Y(n683));
AND2X1_RVT U1479 (.A1(\memory[4][6]), .A2(n733), .Y(n1279));
AND2X1_RVT U1480 (.A1(n1277), .A2(n1280), .Y(n1278));
AND2X1_RVT U1481 (.A1(\memory[4][7]), .A2(n1291), .Y(n1283));
AND2X1_RVT U1482 (.A1(n1281), .A2(n1280), .Y(n1282));
OR2X1_RVT U1483 (.A1(n1283), .A2(n1282), .Y(n680));
AND2X1_RVT U1484 (.A1(\memory[5][3]), .A2(n1409), .Y(n1285));
NBUFFX2_RVT U1485 (.A(n808), .Y(n1402));
AND2X1_RVT U1486 (.A1(data_in_reg[3]), .A2(n1402), .Y(n1284));
AND2X1_RVT U1487 (.A1(\memory[4][2]), .A2(n1288), .Y(n1287));
AND2X1_RVT U1488 (.A1(n1244), .A2(n726), .Y(n1286));
AND2X1_RVT U1489 (.A1(\memory[4][1]), .A2(n733), .Y(n1290));
AND2X1_RVT U1490 (.A1(n1239), .A2(n1292), .Y(n1289));
AND2X1_RVT U1491 (.A1(\memory[4][0]), .A2(n1291), .Y(n1295));
AND2X1_RVT U1492 (.A1(n1293), .A2(n726), .Y(n1294));
OR2X1_RVT U1493 (.A1(n1295), .A2(n1294), .Y(n687));
AND2X1_RVT U1494 (.A1(data_out_reg[4]), .A2(n1353), .Y(n1323));
AND2X1_RVT U1495 (.A1(n1070), .A2(\memory[13][4]), .Y(n1303));
AND2X1_RVT U1496 (.A1(n1354), .A2(\memory[1][4]), .Y(n1302));
AND2X1_RVT U1497 (.A1(n824), .A2(\memory[5][4]), .Y(n1301));
AND2X1_RVT U1498 (.A1(n1356), .A2(\memory[3][4]), .Y(n1299));
AND2X1_RVT U1499 (.A1(n1361), .A2(\memory[7][4]), .Y(n1298));
AND2X1_RVT U1500 (.A1(n1355), .A2(\memory[11][4]), .Y(n1297));

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AND2X1_RVT U1501 ( .A1(n1373), .A2(\memory[9][4] ), .Y(n1296) );
OR4X1_RVT U1502 ( .A1(n1299), .A2(n1298), .A3(n1297), .A4(n1296), .Y(n1300)
    );
OR4X1_RVT U1503 ( .A1(n1303), .A2(n1302), .A3(n1301), .A4(n1300), .Y(n1304)
    );
AND2X1_RVT U1504 ( .A1(n1304), .A2(n1492), .Y(n1319) );
AND2X1_RVT U1505 ( .A1(\memory[15][4] ), .A2(n1305), .Y(n1318) );
AND2X1_RVT U1506 ( .A1(n1371), .A2(\memory[0][4] ), .Y(n1315) );
AND2X1_RVT U1507 ( .A1(n1372), .A2(\memory[4][4] ), .Y(n1314) );
AND2X1_RVT U1508 ( .A1(n825), .A2(\memory[2][4] ), .Y(n1312) );
AND2X1_RVT U1509 ( .A1(n826), .A2(\memory[6][4] ), .Y(n1311) );
AND2X1_RVT U1510 ( .A1(n1373), .A2(\memory[8][4] ), .Y(n1309) );
AND2X1_RVT U1511 ( .A1(n1374), .A2(\memory[12][4] ), .Y(n1308) );
AND2X1_RVT U1512 ( .A1(n1375), .A2(\memory[10][4] ), .Y(n1307) );
AND2X1_RVT U1513 ( .A1(n1376), .A2(\memory[14][4] ), .Y(n1306) );
OR4X1_RVT U1514 ( .A1(n1309), .A2(n1308), .A3(n1307), .A4(n1306), .Y(n1310)
    );
OR3X1_RVT U1515 ( .A1(n1312), .A2(n1311), .A3(n1310), .Y(n1313) );
OR3X1_RVT U1516 ( .A1(n1315), .A2(n1314), .A3(n1313), .Y(n1316) );
AND2X1_RVT U1517 ( .A1(n1469), .A2(n1316), .Y(n1317) );
OR3X1_RVT U1518 ( .A1(n1319), .A2(n1318), .A3(n1317), .Y(n1320) );
AND2X1_RVT U1519 ( .A1(n1321), .A2(n1320), .Y(n1322) );
OR2X1_RVT U1520 ( .A1(n1323), .A2(n1322), .Y(n577) );
AND2X1_RVT U1521 ( .A1(data_out_reg[1]), .A2(n1324), .Y(n1352) );
AND2X1_RVT U1522 ( .A1(n1374), .A2(\memory[13][1] ), .Y(n1332) );
AND2X1_RVT U1523 ( .A1(n1354), .A2(\memory[1][1] ), .Y(n1331) );
AND2X1_RVT U1524 ( .A1(n824), .A2(\memory[5][1] ), .Y(n1330) );
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AND2X1_RVT U1525 ( .A1(n1356), .A2(\memory[3][1] ), .Y(n1328) );
AND2X1_RVT U1526 ( .A1(n1361), .A2(\memory[7][1] ), .Y(n1327) );
AND2X1_RVT U1527 ( .A1(n1355), .A2(\memory[11][1] ), .Y(n1326) );
AND2X1_RVT U1528 ( .A1(n1069), .A2(\memory[9][1] ), .Y(n1325) );
OR4X1_RVT U1529 ( .A1(n1328), .A2(n1327), .A3(n1326), .A4(n1325), .Y(n1329)
    );
OR4X1_RVT U1530 ( .A1(n1332), .A2(n1331), .A3(n1330), .A4(n1329), .Y(n1333)
    );
AND2X1_RVT U1531 ( .A1(n1333), .A2(n1492), .Y(n1349) );
AND2X1_RVT U1532 ( .A1(\memory[15][1] ), .A2(n1370), .Y(n1348) );
AND2X1_RVT U1533 ( .A1(n1371), .A2(\memory[0][1] ), .Y(n1344) );
AND2X1_RVT U1534 ( .A1(n1372), .A2(\memory[4][1] ), .Y(n1343) );
AND2X1_RVT U1535 ( .A1(n1067), .A2(\memory[2][1] ), .Y(n1341) );
AND2X1_RVT U1536 ( .A1(n1068), .A2(\memory[6][1] ), .Y(n1340) );
AND2X1_RVT U1537 ( .A1(n1373), .A2(\memory[8][1] ), .Y(n1338) );
AND2X1_RVT U1538 ( .A1(n1374), .A2(\memory[12][1] ), .Y(n1337) );
AND2X1_RVT U1539 ( .A1(n1375), .A2(\memory[10][1] ), .Y(n1336) );
AND2X1_RVT U1540 ( .A1(n1334), .A2(\memory[14][1] ), .Y(n1335) );
OR4X1_RVT U1541 ( .A1(n1338), .A2(n1337), .A3(n1336), .A4(n1335), .Y(n1339)
    );
OR3X1_RVT U1542 ( .A1(n1341), .A2(n1340), .A3(n1339), .Y(n1342) );
OR3X1_RVT U1543 ( .A1(n1344), .A2(n1343), .A3(n1342), .Y(n1345) );
AND2X1_RVT U1544 ( .A1(n1346), .A2(n1345), .Y(n1347) );
OR3X1_RVT U1545 ( .A1(n1349), .A2(n1348), .A3(n1347), .Y(n1350) );
AND2X1_RVT U1546 ( .A1(n1321), .A2(n1350), .Y(n1351) );
OR2X1_RVT U1547 ( .A1(n1352), .A2(n1351), .Y(n574) );
AND2X1_RVT U1548 ( .A1(data_out_reg[7]), .A2(n1353), .Y(n1393) );
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AND2X1_RVT U1549 ( .A1(n822), .A2(\memory[13][7] ), .Y(n1367) );
AND2X1_RVT U1550 ( .A1(n1354), .A2(\memory[1][7] ), .Y(n1366) );
AND2X1_RVT U1551 ( .A1(n824), .A2(\memory[5][7] ), .Y(n1365) );
AND2X1_RVT U1552 ( .A1(n1355), .A2(\memory[11][7] ), .Y(n1360) );
AND2X1_RVT U1553 ( .A1(n891), .A2(\memory[9][7] ), .Y(n1358) );
AND2X1_RVT U1554 ( .A1(n1356), .A2(\memory[3][7] ), .Y(n1357) );
OR2X1_RVT U1555 ( .A1(n1360), .A2(n1359), .Y(n1363) );
AND2X1_RVT U1556 ( .A1(n1361), .A2(\memory[7][7] ), .Y(n1362) );
OR4X1_RVT U1557 ( .A1(n1367), .A2(n1366), .A3(n1365), .A4(n1364), .Y(n1369)
    );
AND2X1_RVT U1558 ( .A1(n1369), .A2(n1368), .Y(n1390) );
AND2X1_RVT U1559 ( .A1(\memory[15][7] ), .A2(n1370), .Y(n1389) );
AND2X1_RVT U1560 ( .A1(n1371), .A2(\memory[0][7] ), .Y(n1386) );
AND2X1_RVT U1561 ( .A1(n1372), .A2(\memory[4][7] ), .Y(n1385) );
AND2X1_RVT U1562 ( .A1(n1356), .A2(\memory[2][7] ), .Y(n1383) );
AND2X1_RVT U1563 ( .A1(n1361), .A2(\memory[6][7] ), .Y(n1382) );
AND2X1_RVT U1564 ( .A1(n1373), .A2(\memory[8][7] ), .Y(n1380) );
AND2X1_RVT U1565 ( .A1(n1374), .A2(\memory[12][7] ), .Y(n1379) );
AND2X1_RVT U1566 ( .A1(n1375), .A2(\memory[10][7] ), .Y(n1378) );
AND2X1_RVT U1567 ( .A1(n1376), .A2(\memory[14][7] ), .Y(n1377) );
OR4X1_RVT U1568 ( .A1(n1380), .A2(n1379), .A3(n1378), .A4(n1377), .Y(n1381)
    );
OR3X1_RVT U1569 ( .A1(n1383), .A2(n1382), .A3(n1381), .Y(n1384) );
OR3X1_RVT U1570 ( .A1(n1386), .A2(n1385), .A3(n1384), .Y(n1387) );
AND2X1_RVT U1571 ( .A1(n1346), .A2(n1387), .Y(n1388) );
OR3X1_RVT U1572 ( .A1(n1390), .A2(n1389), .A3(n1388), .Y(n1391) );
AND2X1_RVT U1573 ( .A1(n1456), .A2(n1391), .Y(n1392) );
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OR2X1_RVT U1574 (.A1(n1393), .A2(n1392), .Y(n580));
AND2X1_RVT U1575 (.A1(\memory[6][7]), .A2(n731), .Y(n1396));
OR2X1_RVT U1576 (.A1(n1396), .A2(n1395), .Y(n664));
AND2X1_RVT U1577 (.A1(full), .A2(wr_en_reg), .Y(N128));
AND2X1_RVT U1578 (.A1(empty), .A2(rd_en_reg), .Y(N129));
AND2X1_RVT U1579 (.A1(\memory[5][0]), .A2(n1409), .Y(n1398));
INVX0_RVT U1580 (.A(n1401), .Y(n1405));
AND2X1_RVT U1581 (.A1(data_in_reg[0]), .A2(n1405), .Y(n1397));
AND2X1_RVT U1582 (.A1(\memory[5][1]), .A2(n1409), .Y(n1400));
AND2X1_RVT U1583 (.A1(data_in_reg[1]), .A2(n1402), .Y(n1399));
AND2X1_RVT U1584 (.A1(\memory[5][2]), .A2(n1401), .Y(n1404));
AND2X1_RVT U1585 (.A1(data_in_reg[2]), .A2(n1402), .Y(n1403));
OR2X1_RVT U1586 (.A1(n1404), .A2(n1403), .Y(n677));
AND2X1_RVT U1587 (.A1(\memory[5][4]), .A2(n1409), .Y(n1407));
AND2X1_RVT U1588 (.A1(data_in_reg[4]), .A2(n1405), .Y(n1406));
OR2X1_RVT U1589 (.A1(n1486), .A2(n1408), .Y(n1410));
AND2X1_RVT U1590 (.A1(\memory[6][3]), .A2(n1413), .Y(n1417));
AND2X1_RVT U1591 (.A1(n1415), .A2(n1414), .Y(n1416));
OR2X1_RVT U1592 (.A1(n1417), .A2(n1416), .Y(n668));
AND2X1_RVT U1593 (.A1(\memory[7][1]), .A2(n1423), .Y(n1419));
AND2X1_RVT U1594 (.A1(data_in_reg[1]), .A2(n1426), .Y(n1418));
OR2X1_RVT U1595 (.A1(n1419), .A2(n1418), .Y(n662));
AND2X1_RVT U1596 (.A1(\memory[7][2]), .A2(n1423), .Y(n1422));
AND2X1_RVT U1597 (.A1(data_in_reg[2]), .A2(n1430), .Y(n1421));
OR2X1_RVT U1598 (.A1(n1422), .A2(n1421), .Y(n661));
AND2X1_RVT U1599 (.A1(\memory[7][3]), .A2(n1423), .Y(n1425));
AND2X1_RVT U1600 (.A1(data_in_reg[3]), .A2(n1430), .Y(n1424));

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  AND2X1_RVT U1601 ( .A1(\memory[7][4] ), .A2(n734), .Y(n1428) );
  AND2X1_RVT U1602 ( .A1(data_in_reg[4]), .A2(n1426), .Y(n1427) );
  OR2X1_RVT U1603 ( .A1(n1428), .A2(n1427), .Y(n659) );
  AND2X1_RVT U1604 ( .A1(\memory[7][5] ), .A2(n734), .Y(n1433) );
  AND2X1_RVT U1605 ( .A1(n1431), .A2(n1430), .Y(n1432) );
  AND2X1_RVT U1606 ( .A1(count_reg[1]), .A2(n1434), .Y(n1442) );
  AND2X1_RVT U1607 ( .A1(count_reg[1]), .A2(n1435), .Y(n1439) );
  AND2X1_RVT U1608 ( .A1(n1437), .A2(n1436), .Y(n1438) );
  OR2X1_RVT U1609 ( .A1(n1439), .A2(n1438), .Y(n1440) );
  AND2X1_RVT U1610 ( .A1(count_reg[0]), .A2(n1440), .Y(n1441) );
  OR3X1_RVT U1611 ( .A1(n1443), .A2(n1442), .A3(n1441), .Y(n591) );
  OR2X1_RVT U1612 ( .A1(n1444), .A2(wr_ptr_reg[0]), .Y(n1445) );
  AND2X1_RVT U1613 ( .A1(n1446), .A2(n1445), .Y(n588) );
  OR2X1_RVT U1614 ( .A1(n1447), .A2(wr_ptr_reg[2]), .Y(n1448) );
  AND2X1_RVT U1615 ( .A1(n1449), .A2(n1448), .Y(n586) );
  OR2X1_RVT U1616 ( .A1(n1450), .A2(wr_ptr_reg[3]), .Y(n1451) );
  AND2X1_RVT U1617 ( .A1(n1452), .A2(n1451), .Y(n585) );
  OR2X1_RVT U1618 ( .A1(n1456), .A2(n1492), .Y(n1453) );
  AND2X1_RVT U1619 ( .A1(n1454), .A2(n1453), .Y(n584) );
  AND4X1_RVT U1620 ( .A1(n1456), .A2(n1492), .A3(n1455), .A4(n1458), .Y(n1461)
       );
  OR2X1_RVT U1621 ( .A1(n1458), .A2(n1457), .Y(n1459) );
  AND2X1_RVT U1622 ( .A1(n1460), .A2(n1459), .Y(n582) );
  AND2X1_RVT U1623 ( .A1(rd_ptr_reg[3]), .A2(n1460), .Y(n1463) );
  AND2X1_RVT U1624 ( .A1(n1461), .A2(n1468), .Y(n1462) );
  OR2X1_RVT U1625 ( .A1(n1463), .A2(n1462), .Y(n581) );
endmodule
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Finally, our design is functionally correct and passes all simulation cases: Reset, write until full, Overflow detection, read until empty, underflow detection, simultaneous read/write operations, reuse memory addresses cyclically (wrap-around), Hold state, data correctness. **In digital design, FIFO is not just a buffer- it's a bridge that ensures seamless data flow.**